PERFECT UNARY FORMS OVER TOTALLY REAL CUBIC FIELDS: AN EULER BOUND, A REGULATOR SANDWICH, AND EXACT COMPUTATIONS

ABSTRACT. Let K be a totally real cubic field (more generally a number field with $r_1+r_2=3$), let n_K be the number of classes of perfect unary Humbert forms over K modulo homothety and units, and let μ_K be the number of unit–classes of vertices of the *squared sail* $\partial\big(\mathrm{conv}\{q(x)\}+\mathbb{R}^3_{\geq 0}\big)$, where $q(x)=(\sigma_i(x)^2)_i$. We prove three things. First, the Ryshkov polyhedron modulo units is a 2–torus, and Euler's formula gives $\frac{1}{2}\mu_K \leq n_K \leq 2\mu_K$, together with the exact identity $n_K = 2\mu_K - \sum_v(\deg v - 3)$. Second, for every order $\mathfrak{o}$ in a totally real cubic field,

$$\frac{4\,\mathrm{Reg}_{\mathfrak{o}}}{(\log 2D_{\mathfrak{o}})^2} \;\leq\; \mu_K \;\leq\; \frac{4\,\mathrm{Reg}_{\mathfrak{o}}}{(\log 2)^2},$$

by an elementary argument in the geometry of numbers: extreme squares are relative minima of the lattice $\mathfrak{o}$, which are separated in logarithmic space, and Hermite's inequality forces an extreme square near every point of the log–unit torus. The lower bound is sharp up to a constant for the simplest cubic fields. Hence $n_K \asymp \mathrm{Reg}_K$ up to a factor $(\log D_K)^2$, and $\sup n_K = \infty$ as soon as $\mathrm{Reg}_K/(\log D_K)^2$ is unbounded. This holds unconditionally along suitable orders $\mathbb{Z}+p\mathcal{O}_K$, and conditionally on the Cohen–Martinet heuristics for a positive proportion of maximal orders. Third, we compute n_K and μ_K by Voronoi's algorithm in exact arithmetic, with an Euler–characteristic check and a certificate for the unit group. This covers all 4804 totally real cubic fields with $D_K \leq 10^5$ and fields with n_K up to 3534. We find $\mu_K/\mathrm{Reg}_K \approx 0.5$–$0.6$ typically, while Ennola's and Shanks' families sit at $\mu_K = 1$.

Three further results relate perfect forms to other objects. First, relative minima link the squared sail to the Klein sails of the lattice $\mathfrak{o}$: by theorems of Bykovskii and German, every relative minimum is a sail vertex or a special point on a triangular facet. Combined with Euler's formula on the sail tori, this shows that the numbers of vertices and facets of the Klein sails modulo units are also $\asymp \mathrm{Reg}_{\mathfrak{o}}$ up to a factor $(\log D_{\mathfrak{o}})^2$, and that $n_K \leq 2\ell_{\mathfrak{o}}$, where $\ell_{\mathfrak{o}}$ is the number of relative minima modulo units. This is the cubic counterpart of Yasaki's bound $h_d \leq \ell_d$. Second, we determine when Leibak's bound $\gamma_K^m \leq \gamma_m^m D_K$ for the Hermite–Humbert constant is attained. For a real quadratic field this happens exactly when $\mathcal{O}_K$ has a $\mathbb{Z}$–basis u, v with $\mathrm{Nm}(u)+\mathrm{Nm}(v)+\mathrm{Nm}(u-v)=0$, and for a totally real cubic field it forces $2\mathcal{O}_K = \mathfrak{pq}^2$. Third, an integral perfect form whose minimal vectors are units is the cubic lattice I_3, and its functional generates the codifferent. Combined with Euler's formula on the totally positive sail, this shows that in no totally real cubic field are all additively indecomposable integers squares of units.

1. INTRODUCTION

1.1. **The problem.** Fix a number field K of degree $n=[K:\mathbb{Q}]$ with r_1 real and r_2 complex places, and let $d := r_1+r_2$. Voronoi's reduction theory of positive definite quadratic forms over $\mathbb{Q}$ [25] was extended to arithmetic minima over number fields by Koecher, Ash, Coulangeon, Okuda–Yano and Watanabe–Yano–Hayashi. In the unary (1×1) case the objects are positive *Humbert forms* $a=(a_\sigma)_\sigma \in P_1(K_\mathbb{R}) = \prod_{\sigma|\infty} P_1(K_\sigma)$, and one studies the finite set of *perfect* unary forms up to homothety and the action of the unit group. Write

$$n_K \;:=\; \#\big(\partial^0 K_1(m_{\mathfrak{o}})/\Gamma_{\mathfrak{o}}\big)$$

for the number of these classes (notation recalled in §2). The question motivating this note is: *how large is n_K, and in particular what can be proved from below?*

Date: October 28, 2026.

1.2. **Known cases.** For real quadratic fields ($d = 2$), Yasaki [27] bounds the number h_d of classes of perfect unary forms above and below in terms of the period of an associated continued fraction [27, Thm. 5.1]. He shows that $h_d = 1$ for the fields $\mathbb{Q}(\sqrt{d})$ with $d = (2k+1)^2 + 1$ squarefree, and that h_d is unbounded. Leibak, Porter and Ling [18] determine all real quadratic fields with $h_d = 1$, proving the four families found by Yasaki's computations for $d < 200000$ [27, Remark after Cor. 5.1]. The first cubic case is due to Komatsu and Watanabe [17]: for $\ell \geq 2$ with $4\ell^3+27$ squarefree, the cubic field $\mathbb{Q}(v)$, $v^3+\ell v-1 = 0$, has a single real place ($r_1 = r_2 = 1$, so $d = 2$), and over it there is exactly one perfect unary form up to homothety and units, i.e. $n_K = 1$. For arbitrary totally real K, Porter and Mendelsohn [21] give an upper bound in terms of the additive Hermite–Humbert constant and the covering radius of the log–unit lattice. For $d \geq 3$ the other systematic inputs are the construction of perfect forms by Gunnells–Yasaki [10] and the finiteness theorem of Okuda–Yano [20].

1.3. **Results.** After recalling the framework (§2) we prove the following.

- **Squared–sail duality** (Lemma 3.1, Proposition 3.4). Perfect forms are the facets of the squared sail, and the facets of the Ryshkov polyhedron are its vertices, the *extreme squares*. Perfection is affine independence of squared minimal vectors in the supporting hyperplane; on that hyperplane linear and affine independence coincide.
- **The Euler bound** (Theorem 4.3). If $r_1 + r_2 = 3$, the quotient $\partial K_1(m_\mathfrak{o})/\Gamma_\mathfrak{o}$ is a 2–torus, $\frac{1}{2}\mu_K \leq n_K \leq 2\mu_K$, and $n_K = 2\mu_K - \sum_v (\deg v - 3)$.
- **The regulator sandwich** (Theorem 6.5). For totally real cubic K, $4\operatorname{Reg}_K/(\log 2D_K)^2 \leq \mu_K \leq 4\operatorname{Reg}_K/(\log 2)^2$. The consequences for $\sup n_K$ are in Corollary 6.8, and the analogue in all degrees is (5).
- **Exact computation** (§5). Voronoi's algorithm in exact arithmetic, with two certificates, applied to all totally real cubic fields with $D_K \leq 10^5$ and to large–regulator examples.
- **Klein sails** (§7). Extreme squares are sail vertices or German points (Corollary 7.3); the sails modulo units have $\asymp \operatorname{Reg}_\mathfrak{o}$ vertices and facets up to $(\log D_\mathfrak{o})^2$ (Theorem 7.4); and $n_K \leq 2\ell_\mathfrak{o}$ is the cubic analogue of Yasaki's $h_d \leq \ell_d$ (Remark 4.4).
- **Leibak's bound** (§8). Equality $\gamma_K^m = \gamma_m^m D_K$ is characterised for $m = 2, 3$ (Theorem 8.1, Corollary 8.2).
- **Indecomposables** (§9). Every totally real cubic field has an indecomposable that is not the square of a unit (Theorem 9.4). The proof combines the sail torus with the fact that an integral perfect form with unit minimal vectors is I_3 (Lemma 9.2). Minimal triples of relative minima are bases or degenerate (Lemma 7.8), and we conjecture that $\ell_{\mathcal{O}_K} = 1$ only for monogenic simplest cubic fields (Conjecture 9.8).

Conventions. K_σ denotes the completion of K at the archimedean place σ, with $e_\sigma := [K_\sigma : \mathbb{R}] \in \{1, 2\}$. We embed $K \hookrightarrow K_\mathbb{R} = \prod_\sigma K_\sigma$ by $\iota(\lambda) = (\sigma(\lambda))_\sigma$, fixing one embedding per place. $\mathfrak{o}$ denotes an order in K; the reader may take $\mathfrak{o} = \mathcal{O}_K$. "Form" always means *unary* Humbert form. We write $\langle \cdot, \cdot \rangle$ for the standard inner product on $\mathbb{R}^d$.

2. The Ryshkov polyhedron and the squared embedding

2.1. **Forms, minima, perfection.** For a real place σ the cone $P_1(K_\sigma)$ is $\mathbb{R}_{>0}$; for a complex place it is the cone of positive 1×1 Hermitian matrices, again $\mathbb{R}_{>0}$. Hence

$$P_1(K_\mathbb{R}) = \mathbb{R}^d_{>0}, \qquad d = r_1 + r_2,$$

the open positive orthant indexed by the archimedean places. For $a = (a_\sigma) \in P_1(K_\mathbb{R})$ the arithmetic minimum and the set of shortest vectors are

$$m_\mathfrak{o}(a) = \min_{0 \neq x \in \mathfrak{o}} \sum_\sigma e_\sigma \, a_\sigma \, |\sigma(x)|^2, \qquad S_\mathfrak{o}(a) = \Big\{ x \in \mathfrak{o} : \ \textstyle\sum_\sigma e_\sigma a_\sigma |\sigma(x)|^2 = m_\mathfrak{o}(a) \Big\}.$$

Introduce the *squared embedding*

$$q : \mathfrak{o} \longrightarrow \mathbb{R}^d_{\geq 0}, \qquad q(x) = \big(e_\sigma \, |\sigma(x)|^2 \big)_\sigma, \tag{1}$$

so that $\sum_\sigma e_\sigma a_\sigma |\sigma(x)|^2 = \langle a, q(x) \rangle$ and $m_\mathfrak{o}(a) = \min_{0 \neq x} \langle a, q(x) \rangle$. Since each σ is injective, $q(x)$ has *strictly positive* coordinates for $x \neq 0$. A form a is $\mathfrak{o}$*–perfect* if $\{ q(x) : x \in S_\mathfrak{o}(a) \}$ spans $\mathbb{R}^d$ over $\mathbb{R}$.

2.2. **The Ryshkov polyhedron.** Following Ryshkov [22] and Watanabe–Yano–Hayashi [26], set

$$K_1(m_\mathfrak{o}) = \{ a \in P_1(K_\mathbb{R}) : m_\mathfrak{o}(a) \geq 1 \} = \bigcap_{0 \neq x \in \mathfrak{o}} \{ a \in \mathbb{R}^d_{>0} : \ \langle a, q(x) \rangle \geq 1 \}.$$

We collect the structural facts we use. The discrete group acting is $\Gamma_\mathfrak{o} = \{ g \in \mathrm{GL}_1(K_\mathbb{R}) : g\mathfrak{o} = \mathfrak{o} \} = \mathfrak{o}^\times$, acting by $a \cdot u = (|\sigma(u)|^2 a_\sigma)_\sigma$; this preserves $m_\mathfrak{o}$ and hence $K_1(m_\mathfrak{o})$.

Theorem 2.1 (Ryshkov; Watanabe–Yano–Hayashi; Okuda–Yano)**.** *The set $K_1(m_\mathfrak{o})$ is a closed convex* locally finite *polyhedron of full dimension d in $P_1(K_\mathbb{R})$. Its set of vertices $\partial^0 K_1(m_\mathfrak{o})$ equals the set of perfect forms with $m_\mathfrak{o} = 1$. Any two vertices are joined by a finite edge–path along 1–faces. Finally, $\Gamma_\mathfrak{o}$ acts on the face complex of $K_1(m_\mathfrak{o})$ with finitely many orbits of faces in each dimension.*

Proof. Local finiteness, the identification of the vertices with the perfect forms of minimum 1, and edge–connectivity are [17, Thm. 2.2], proved by Watanabe–Yano–Hayashi [26]; the identification of the vertices is [26, Thm. 2.4]. The finiteness of $\partial^0 K_1(m_\mathfrak{o})/\Gamma_\mathfrak{o}$ is Okuda–Yano [20, Thm. 5.1] (quoted in [17, §1]); together with local finiteness it gives finitely many orbits of faces in each dimension. The theorem numbers of [26, 20] are as quoted in [17]. □

Definition 2.2. Let K have $d = r_1 + r_2$. We set

$$n_K := \#\big(\partial^0 K_1(m_\mathfrak{o}) / \Gamma_\mathfrak{o} \big), \qquad \mu_K := \#\big(\{ \text{facets of } K_1(m_\mathfrak{o}) \} / \Gamma_\mathfrak{o} \big).$$

Thus n_K counts perfect forms up to homothety and units, and μ_K counts classes of *extreme squares*, i.e. of minimal vectors x for which $q(x)$ is a vertex of the squared sail (Proposition 3.4); in general not every minimal vector is extreme. Both are finite by Theorem 2.1.

3. The affine criterion and the squared sail

3.1. **An affine perfection criterion.** The definition of perfection asks the squared vectors $q(x)$, $x \in S_\mathfrak{o}(a)$, to span $\mathbb{R}^d$ *linearly*. The next lemma replaces this by the more transparent — and dimension–correct — condition of *affine* independence.

Lemma 3.1. *Let $a \in P_1(K_\mathbb{R})$ and put $m = m_\mathfrak{o}(a) > 0$. All squared minimal vectors lie on the affine hyperplane*

$$\Pi_a = \{ w \in \mathbb{R}^d : \langle a, w \rangle = m \}, \qquad 0 \notin \Pi_a.$$

Consequently the following are equivalent:

(i) *a is $\mathfrak{o}$–perfect, i.e. $\{ q(x) : x \in S_\mathfrak{o}(a) \}$ spans $\mathbb{R}^d$;*
(ii) *$\{ q(x) : x \in S_\mathfrak{o}(a) \}$ contains d affinely independent points;*
(iii) *$\{ q(x) : x \in S_\mathfrak{o}(a) \}$ affinely spans Π_a.*

In particular a perfect form has at least d minimal vectors up to roots of unity. On Π_a, which misses the origin, linear and affine independence of the $q(x)$ coincide*; the content of the lemma is the geometric statement that perfection means the squares affinely fill the supporting hyperplane. What is* not *equivalent to perfection is linear independence of the underlying field elements x, because the map q is non–linear.*

Proof. By definition every $x \in S_{\mathfrak{o}}(a)$ satisfies $\langle a, q(x)\rangle = m$, so $q(x) \in \Pi_a$. As $m > 0$ we have $0 \notin \Pi_a$, and Π_a is an affine hyperplane of dimension $d-1$.

(iii)⇒(i): if a subset $S \subseteq \Pi_a$ affinely spans the $(d-1)$–flat Π_a, then its linear span contains the linear span of Π_a. Since Π_a is a hyperplane not through the origin, its linear span is all of $\mathbb{R}^d$; hence S spans $\mathbb{R}^d$.

(i)⇒(iii): suppose the affine span of S were a proper flat $\Pi' \subsetneq \Pi_a$, of dimension $\leq d-2$. Then $\operatorname{span}_{\mathbb{R}} S \subseteq \operatorname{span}_{\mathbb{R}} \Pi'$, which has dimension $\leq d-1$, contradicting $\operatorname{span}_{\mathbb{R}} S = \mathbb{R}^d$.

(ii)⇔(iii): a subset of a $(d-1)$–flat affinely spans it iff it contains d affinely independent points (an affine basis of Π_a has d elements).

The final sentence is immediate from (ii): d affinely independent points require at least d distinct $q(x)$; since $q(\zeta x) = q(x)$ exactly for roots of unity ζ (Lemma 3.3 (a) below), these are d distinct classes. Finally, on Π_a linear and affine independence agree: if $\sum_i c_i\, q(x_i) = 0$ is a linear relation among points of Π_a, pairing with a gives $m \sum_i c_i = 0$, whence $\sum_i c_i = 0$ as $m \neq 0$, which is an affine relation; the converse is trivial. Thus a subset of d of the squares is linearly independent in $\mathbb{R}^d$ iff it is affinely independent in Π_a, and either detects perfection. By contrast the underlying field elements $x_1, \ldots, x_d$ may be linearly independent over $\mathbb{Q}$ while $q(x_1), \ldots, q(x_d)$ are affinely dependent, and conversely; this is why linear independence of the x is not the right criterion. □

Remark 3.2. Lemma 3.1 clarifies a subtlety in passing from a set of m linearly independent minimal vectors $\eta_1, \ldots, \eta_m$ to perfection: what must be independent are not the η_i but the squares $q(\eta_i)$, and the relevant independence is affine in the hyperplane Π_a. For $d = 2$ (real quadratic, complex cubic) the hyperplane Π_a is a line, and any two distinct points on it are affinely independent; so the phenomenon first has content at $d = 3$.

3.2. **The squared sail and polar duality.** Let $Q := q(\mathfrak{o} \setminus \{0\}) \subset \mathbb{R}^d_{>0}$ and define the *squared–sail body* and *sail*

$$C := \operatorname{conv}(Q) + \mathbb{R}^d_{\geq 0}, \qquad \mathcal{S}^{\sqsupset} := \partial C.$$

Because every $q(x)$ has positive coordinates and the lattice $\iota(\mathfrak{o})$ is discrete, C is a closed convex set with nonempty interior and recession cone $\mathbb{R}^d_{\geq 0}$; $\mathcal{S}^{\sqsupset}$ is the part of ∂C visible from the origin. We record the basic equivariance.

Lemma 3.3. *For $u \in \mathfrak{o}^\times$ let $D_u = \operatorname{diag}(|\sigma(u)|^2)_\sigma$ act on $\mathbb{R}^d$. Then:*

(a) *$q(ux) = D_u\, q(x)$ and $q(\zeta x) = q(x)$ for every root of unity $\zeta \in \mathfrak{o}^\times$; moreover $D_u = \mathrm{Id}$ iff u is a root of unity.*

(b) *D_u preserves Q, C, $\mathcal{S}^{\sqsupset}$ and the cone $\mathbb{R}^d_{>0}$, and is dual to the $\Gamma_{\mathfrak{o}}$–action on forms: $\langle a \cdot u,\, q(x)\rangle = \langle a,\, q(ux)\rangle$.*

Proof. (a) From (1), $q(ux)_\sigma = e_\sigma |\sigma(u)|^2 |\sigma(x)|^2 = (D_u q(x))_\sigma$. If u is a root of unity then $|\sigma(u)| = 1$ for all σ, so $D_u = \mathrm{Id}$ and $q(\zeta x) = q(x)$. Conversely $D_u = \mathrm{Id}$ forces $|\sigma(u)| = 1$ for every archimedean σ; by Kronecker's theorem u is a root of unity. (b) is immediate since D_u is a positive diagonal map permuting $Q = q(\mathfrak{o}^\times \cdot \mathfrak{o})$ and fixing $\mathbb{R}^d_{\geq 0}$, and $\langle a \cdot u, q(x)\rangle = \sum_\sigma e_\sigma |\sigma(u)|^2 a_\sigma |\sigma(x)|^2 = \langle a, q(ux)\rangle$. □

Proposition 3.4 (Squared–sail duality). *With the notation above,*

$$K_1(m_{\mathfrak{o}}) = \{a \in \mathbb{R}^d_{>0} : \ \langle a, w\rangle \geq 1 \text{ for all } w \in C\}.$$

The face lattices of $K_1(m_{\mathfrak{o}})$ and of the sail $\mathcal{S}^{\square}$ are anti–isomorphic; explicitly:

(i) *the facets of $K_1(m_{\mathfrak{o}})$ are exactly the sets $K_1(m_{\mathfrak{o}}) \cap \{\langle a, w\rangle = 1\}$ for w a vertex of C (equivalently, a vertex of $\mathcal{S}^{\square}$, equivalently an extreme point of $\operatorname{conv} Q$); distinct vertices give distinct facets;*

(ii) *the vertices of $K_1(m_{\mathfrak{o}})$ — the perfect forms — are exactly the facets of $\mathcal{S}^{\square}$.*

*Consequently μ_K equals the number of $\mathfrak{o}^\times$–classes of vertices of $\mathcal{S}^{\square}$ (*extreme squares*), and n_K equals the number of $\mathfrak{o}^\times$–classes of facets of $\mathcal{S}^{\square}$.*

Proof. For $a \in \mathbb{R}^d_{>0}$ we have $\langle a, w\rangle \geq 1$ for all $w \in Q$ iff for all $w \in \operatorname{conv} Q$ (the constraint is affine in w) iff for all $w \in C$ (adding any $v \in \mathbb{R}^d_{\geq 0}$ only increases $\langle a, w\rangle$ since $a > 0$). This proves the displayed identity, exhibiting $K_1(m_{\mathfrak{o}})$ as the polar dual (intersected with the open cone) of the closed convex body C with recession cone $\mathbb{R}^d_{\geq 0}$.

By Theorem 2.1, $K_1(m_{\mathfrak{o}})$ is a locally finite polyhedron, so polarity restricts to a face–wise anti–isomorphism between the bounded faces of $K_1(m_{\mathfrak{o}})$ and the faces of C supported by functionals $a \in \mathbb{R}^d_{>0}$ (the "lower", origin–facing faces, i.e. the faces of $\mathcal{S}^{\square}$), by standard polarity for polyhedra. Under this anti–isomorphism a facet (codimension–1 face) of $K_1(m_{\mathfrak{o}})$ corresponds to a 0–face (vertex) of C, and a 0–face (vertex) of $K_1(m_{\mathfrak{o}})$ corresponds to a facet of C. A facet of $K_1(m_{\mathfrak{o}})$ is cut out by an *irredundant* inequality $\langle a, w\rangle \geq 1$; irredundancy means precisely that w is an extreme point of C not dominated by a convex combination of the others, i.e. a vertex of C, which because $\mathbb{R}^d_{\geq 0}$ is the recession cone is a vertex of $\operatorname{conv} Q$ lying on $\mathcal{S}^{\square}$. Distinct vertices $w \neq w'$ produce distinct facet normals, hence distinct facets. This is (i); (ii) is the dual statement, using that perfect forms are the vertices of $K_1(m_{\mathfrak{o}})$ (Theorem 2.1). The last sentence follows by passing to $\Gamma_{\mathfrak{o}}$–orbits via Lemma 3.3(b) and Definition 2.2. □

Thus "count perfect forms" (n_K) and "count extreme squares" (μ_K) are the two natural enumerative invariants attached to the same sail; the next section relates them.

4. The Euler bound

Throughout this section K is a number field with $d = r_1 + r_2 = 3$. The signatures are $(r_1, r_2) = (3,0), (2,1), (1,2), (0,3)$, i.e. totally real cubic, mixed quartic, quintic with one real place, and totally complex sextic fields; the unit rank is 2 in every case. We first set up a uniform logarithmic chart.

Lemma 4.1. *Let $H_0 = \{t \in \mathbb{R}^d : \sum_\sigma t_\sigma = 0\}$ and let $\mathbf{e} = (e_\sigma)_\sigma$. The map*

$$\Phi : \mathbb{R}^d_{>0} \to \mathbb{R}^d, \qquad \Phi(a)_\sigma = e_\sigma \log a_\sigma,$$

is a homeomorphism carrying the radial scaling $a \mapsto ta$ to the translation $t \mapsto t + (\log t)\,\mathbf{e}$, and carrying the $\mathfrak{o}^\times$–action $a \mapsto a \cdot u$ to the translation

$$t \mapsto t + 2\,L(u), \qquad L(u) := \big(e_\sigma \log|\sigma(u)|\big)_\sigma \in H_0.$$

The set $L(\mathfrak{o}^\times) \subset H_0$ is a full lattice of rank $d - 1 = 2$ (Dirichlet's unit theorem), with covolume in H_0 equal to the regulator Reg_K up to the fixed normalizing constant of H_0. Finally $\mathbf{e} \notin H_0$ (since $\sum_\sigma e_\sigma = n > 0$), so $\mathbb{R}^d = H_0 \oplus \mathbb{R}\mathbf{e}$.

Proof. Φ is a homeomorphism with inverse $t \mapsto (e^{t_\sigma/e_\sigma})_\sigma$. The scaling statement is $\Phi(ta)_\sigma = e_\sigma \log(ta_\sigma) = \Phi(a)_\sigma + e_\sigma \log t$. For the unit action, $\Phi(a \cdot u)_\sigma = e_\sigma \log(|\sigma(u)|^2 a_\sigma) = \Phi(a)_\sigma + 2e_\sigma \log|\sigma(u)|$. We have $\sum_\sigma e_\sigma \log|\sigma(u)| = \log \prod_\sigma |\sigma(u)|^{e_\sigma} = \log|\operatorname{Nm}(u)| = 0$, so $L(u) \in H_0$,

and $u \mapsto L(u)$ is the standard Dirichlet logarithmic embedding, whose image is a lattice of rank $r_1 + r_2 - 1 = 2$ with covolume Reg_K. The decomposition is clear. □

Lemma 4.2. *$\partial K_1(m_{\mathfrak{o}}) = \{a \in \mathbb{R}^d_{>0} : m_{\mathfrak{o}}(a) = 1\}$ is homeomorphic to $\mathbb{R}^2$, and projection along $\mathbb{R}\mathbf{e}$ gives a homeomorphism $\pi : \partial K_1(m_{\mathfrak{o}}) \xrightarrow{\sim} H_0$ that is equivariant for the $\mathfrak{o}^\times$–action on the source and the translation action $t \mapsto t + 2L(u)$ on H_0.*

Proof. The function $a \mapsto m_{\mathfrak{o}}(a) = \min_{x \neq 0} \langle a, q(x) \rangle$ is concave (a pointwise minimum of linear functions), positively homogeneous of degree 1, and strictly positive on $\mathbb{R}^d_{>0}$: indeed for fixed $a \gg 0$ each $\langle a, q(x) \rangle > 0$, and the infimum is attained and positive because $\iota(\mathfrak{o})$ is a full discrete lattice and $\langle a, q(\cdot) \rangle$ is an anisotropic positive form on it. Hence each ray $\mathbb{R}_{>0} \cdot a$ meets the level set $\{m_{\mathfrak{o}} = 1\}$ in exactly one point, so radial projection identifies $\{m_{\mathfrak{o}} = 1\}$ with the cross–section $\mathbb{R}^d_{>0}/\mathbb{R}_{>0}$. In the chart Φ of Lemma 4.1 the radial direction is $\mathbb{R}\mathbf{e}$, transverse to H_0; thus $\mathbb{R}^d_{>0}/\mathbb{R}_{>0} \cong H_0 \cong \mathbb{R}^2$, and the composite $\pi = (\text{project along } \mathbb{R}\mathbf{e}) \circ \Phi$ is a homeomorphism $\{m_{\mathfrak{o}} = 1\} \to H_0$. It is equivariant because the unit action is the translation $t \mapsto t + 2L(u)$ with $2L(u) \in H_0$ (Lemma 4.1), which commutes with projection along $\mathbb{R}\mathbf{e}$. □

Theorem 4.3 (Euler bound)**.** *Let K be a number field with $r_1 + r_2 = 3$ (e.g. any totally real cubic field), and let $\mathfrak{o}$ be an order in K. Then $\partial K_1(m_{\mathfrak{o}})/\Gamma_{\mathfrak{o}}$ is a 2–torus, and*

$$\boxed{\tfrac{1}{2}\,\mu_K \;\le\; n_K \;\le\; 2\,\mu_K}$$

where n_K, μ_K are as in Definition 2.2. Moreover:

(a) *$n_K = 2\mu_K$ if and only if every perfect form is* simplicial*, i.e. its dual polygon* $\mathrm{conv}\{q(x) : x \in S_{\mathfrak{o}}(a)\}$ *is a triangle (equivalently, every vertex of $K_1(m_{\mathfrak{o}})$ lies on exactly three facets). More precisely,*

$$n_K \;=\; 2\mu_K - \sum_{v} \big(\deg v - 3\big), \tag{2}$$

the sum running over the vertex classes;

(b) *$n_K = \frac{1}{2}\mu_K$ if and only if every facet of $K_1(m_{\mathfrak{o}})$ is a triangle.*

Proof. By Lemma 4.2, $X := \partial K_1(m_{\mathfrak{o}}) \cong \mathbb{R}^2$, and $\Gamma_{\mathfrak{o}} = \mathfrak{o}^\times$ acts on X through the translation action of $L(\mathfrak{o}^\times)$ on $H_0 \cong \mathbb{R}^2$. By Lemma 3.3(a) the kernel of this action is exactly the group of roots of unity, so the effective action is by $\mathfrak{o}^\times/\mu(K) \cong \mathbb{Z}^2$ (Lemma 4.1) and is *free*; it is properly discontinuous and cocompact because $L(\mathfrak{o}^\times)$ is a full lattice in H_0 of covolume $\mathrm{Reg}_K < \infty$. A free, properly discontinuous, cocompact action of $\mathbb{Z}^2$ on $\mathbb{R}^2$ has quotient a closed aspherical surface with fundamental group $\mathbb{Z}^2$, namely the 2–torus T; in particular $\chi(T) = 0$.

The polyhedral face complex of $K_1(m_{\mathfrak{o}})$ restricts to X as a locally finite regular CW decomposition (Theorem 2.1), invariant under $\Gamma_{\mathfrak{o}}$ and with finitely many orbits of cells in each dimension (Theorem 2.1). As the action is free, this descends to a finite CW decomposition of T with

$$V = n_K \;\; \text{(0–cells: perfect forms)}, \quad E = e_K \;\; \text{(1–cells: edges)}, \quad F = \mu_K \;\; \text{(2–cells: facets)},$$

the identification $F = \mu_K$ being Proposition 3.4 and Definition 2.2. To see that the descent produces exactly these cell counts with no folding, note that no cell is set–stabilized by a nontrivial unit. Indeed each facet of $K_1(m_{\mathfrak{o}})$ is supported by some $q(x)$ and is cut from $\mathbb{R}^d_{>0}$ by an equation $\sum_\sigma e_\sigma a_\sigma |\sigma(x)|^2 = 1$ with strictly positive coefficients, hence is contained in the bounded simplex $\{a \ge 0 : \sum_\sigma e_\sigma a_\sigma |\sigma(x)|^2 = 1\}$; each facet, and therefore each of its edges and vertices, is a bounded set. The unit group acts on the logarithmic chart H_0 by the translations $L(u)$, and a nonzero translation cannot preserve a bounded set. Thus no facet, edge, or vertex is fixed setwise

by a nontrivial $u \in \mathfrak{o}^\times/\mu(K)$, the 2–cells of T are genuine disks and its 0–cells genuine points, and the quotient has exactly μ_K two–cells, e_K one–cells, and n_K zero–cells. Euler's formula gives

$$V - E + F = \chi(T) = 0. \tag{3}$$

Now use the polytopal structure of $K_1(m_\mathfrak{o})$, a 3–dimensional convex polyhedron. Each vertex (a 0–face of a 3–polyhedron) is incident to at least three edges and three facets; hence in the surface X every vertex has degree ≥ 3, and the same holds for the 0–cells of T. Each facet (a 2–face) is a convex polygon, hence is bounded by at least three edges; the same holds for the 2–cells of T. Counting edge–incidences in T two ways,

$$2E = \sum_v \deg(v) \;\geq\; 3V, \qquad 2E = \sum_f (\#\text{sides of } f) \;\geq\; 3F. \tag{4}$$

From (3), $E = V + F$. Substituting into the two inequalities of (4):

$$2(V+F) \geq 3V \implies F \geq \tfrac{1}{2}V \implies V \leq 2F,$$

$$2(V+F) \geq 3F \implies V \geq \tfrac{1}{2}F \implies V \geq \tfrac{1}{2}F.$$

Together, $\frac{1}{2}F \leq V \leq 2F$, i.e. $\frac{1}{2}\mu_K \leq n_K \leq 2\mu_K$.

For (a): the upper bound $V \leq 2F$ is an equality iff $2(V+F) = 2E = 3V$, i.e. iff every vertex has degree exactly 3. A degree–3 vertex of $K_1(m_\mathfrak{o})$ lies on exactly three facets, i.e. its dual polygon has three vertices (Proposition 3.4); this is the definition of simplicial. Identity (2) is $2E = \sum_v \deg v$ combined with $E = V + F$. (When all vertices have degree 3 one has $E = \frac{3}{2}V$ and then $F = \frac{1}{2}V$, forcing the average face to be a hexagon: the hexagonal tiling of the torus.) For (b): the lower bound $V \geq \frac{1}{2}F$ is an equality iff $2E = 3F$, i.e. iff every facet is a triangle. In the totally real case a vertex has degree at most 6, because $x \mapsto \mathrm{Tr}(\alpha x^2)$ is a positive definite ternary form and so has at most 6 minimal vectors up to sign. Hence equality in (b) forces every perfect form to have six extreme minimal vectors up to sign. □

Remark 4.4. In the real quadratic case $d = 2$ the quotient $\partial K_1/\mathfrak{o}^\times$ is a circle, $\chi = 0$ gives $V = E$, and the cells are indexed by a continued fraction; this is the setting of [27]. There $\ell_d/2 \leq h_d \leq \ell_d$ [27, Thm. 5.1], where ℓ_d, the period length of the continued fraction, is the number of relative minima modulo units. Theorem 4.3 is the two–dimensional analogue of the Euler part of this picture. The analogue of the period is not μ_K but the number $\ell_\mathfrak{o}$ of relative minima modulo units (§6). These are the 2–cells of the Minkowski–Voronoi complex, a second torus with f–vector $(2\ell_\mathfrak{o}, 3\ell_\mathfrak{o}, \ell_\mathfrak{o})$ (Proposition 7.7), and Theorem 4.3 with Lemma 6.3 gives $n_K \leq 2\mu_K \leq 2\ell_\mathfrak{o}$. In degree 2 the Minkowski–Voronoi complex is a circle with ℓ_d vertices, so in both degrees n_K is at most the number of its vertices. The census gives $\frac{4}{9}\ell_\mathfrak{o} \leq n_K$ for all $D_K \leq 10^5$ (Observation 5.7); a proof of a lower bound $n_K \geq c\,\ell_\mathfrak{o}$ would complete the analogy (Problem (4)). As in the quadratic case, where ℓ_d and $\log \varepsilon_d$ agree up to a factor $\log d$, μ_K and $\ell_\mathfrak{o}$ are comparable to Reg_K up to a factor $(\log D_K)^2$ (Theorem 6.5).

Corollary 4.5. *For K with $r_1 + r_2 = 3$ one always has $\mu_K \geq 1$, hence $n_K \geq 1$; and $n_K \leq 2\mu_K$. If $\mu_K = 1$ then $n_K \in \{1, 2\}$. The value $(\mu_K, n_K) = (1, 2)$ is realized by a single hexagonal 2–cell on T ($V = 2, E = 3, F = 1$).*

Proof. $K_1(m_\mathfrak{o})$ is full–dimensional and unbounded with at least one vertex (a perfect form exists, Theorem 2.1), hence has at least one facet; so $\mu_K \geq 1$. The remaining assertions are the bound of Theorem 4.3 with $F = 1$: then $E = V + 1$ and $2(V+1) \geq 3V$ give $V \leq 2$, while $2E \geq 3F = 3$ gives $E \geq 2$, so $V \geq 1$. The realizing complex is the standard one–face hexagonal torus. □

Proposition 4.6. *Let K be totally real cubic and $\mathfrak{o}$ an order with $n_K = 1$. Then either*

(a) $\mu_K = 1$, *the unique vertex class has degree* 4, *and* T *consists of one quadrilateral* ($V = 1, E = 2, F = 1$)*; or*
(b) $\mu_K = 2$, *the unique vertex class has degree* 6, *and* T *consists of two triangles* ($V = 1, E = 3, F = 2$).

In case (b) *the trace lattice* $(\mathfrak{o}, \mathrm{Tr}(\alpha xy))$ *of the perfect form is similar to the face–centred cubic lattice* A_3, *so* $\gamma_K^3 = 2D_{\mathfrak{o}}$*; if* $\mathfrak{o} = \mathcal{O}_K$, *then* $2\mathcal{O}_K = \mathfrak{pq}^2$ *by Theorem 8.1.*

Proof. With $V = 1$, identity (2) reads $1 = 2\mu_K - (\delta - 3)$, where δ is the degree of the vertex. The degree is the number of extreme minimal vectors up to sign, so $3 \le \delta \le 6$ (proof of Theorem 4.3). Hence $\delta = 2\mu_K + 2$ gives $(\mu_K, \delta) \in \{(1,4), (2,6)\}$, with $E = V + F$. In case (b) the positive ternary form $\mathrm{Tr}(\alpha x^2)$ has 12 minimal vectors. The only 3–dimensional lattice with 12 minimal vectors is A_3, up to similarity, so the form attains γ_3. Its minimum is 1 and its determinant is $\mathrm{Nm}(\alpha) D_{\mathfrak{o}}$, so $\gamma_K^3 \ge 1/\mathrm{Nm}(\alpha) = \gamma_3^3 D_{\mathfrak{o}}$. Hermite's inequality gives the reverse. □

No totally real cubic field with $D_K \le 10^5$ has $n_K = 1$ (Observation 5.7).

5. Exact computations

We begin with Ennola's totally real cubic fields [8]

$$K = \mathbb{Q}(\rho), \qquad f_a(x) = x^3 + (a-1)x^2 - ax - 1 = 0, \qquad a \ge 2,$$

which are totally real with $\mathbb{Z}[\rho] \subseteq \mathcal{O}_K$ and (for $a \ge 3$) fundamental units $\rho, \rho - 1$; the member $a = 2$ is the cyclic cubic field of conductor 7 and discriminant 49. These are "simplest"–type families in the sense of Shanks [23] and Lemmermeyer–Pethő [19].

5.1. **An exact norm identity.** The following elementary identity governs which non–unit elements can reach the sail.

Lemma 5.1. *For* $K = \mathbb{Q}(\rho)$ *as above,*

$$\mathrm{Nm}(\rho) = 1, \qquad \mathrm{Nm}(\rho - 1) = 1, \qquad \mathrm{Nm}(1 + \rho) = 3 - 2a.$$

In particular ρ *and* $\rho - 1$ *are units, while* $1 + \rho$ *has norm* $3 - 2a$*. For* $a \ge 3$ *the least absolute norm of a non–unit of the form* $\rho - c$ ($c \in \mathbb{Z}$) *is* $2a - 3$, *attained at* $c = -1$*. The case* $a = 2$ *is exceptional: there* $|3 - 2a| = 1$, *so* $1 + \rho$ *is itself a unit and* $K = \mathbb{Q}(\zeta_7)^+$ *has discriminant* 49*; the rational primes* 2, 3, 5 *are inert, and the least non–unit norm is* 7, *attained at* $\mathrm{Nm}(2 - \rho) = -7$.

Proof. Writing $f_a(x) = \prod_i (x - \rho_i)$ over the conjugates, for $c \in \mathbb{Z}$ we have $\mathrm{Nm}(\rho - c) = \prod_i (\rho_i - c) = (-1)^3 \prod_i (c - \rho_i) = -f_a(c)$. Hence

$$\mathrm{Nm}(\rho) = -f_a(0) = -(-1) = 1, \quad \mathrm{Nm}(\rho - 1) = -f_a(1) = -(1 + (a-1) - a - 1) = 1,$$

$$\mathrm{Nm}(1 + \rho) = \mathrm{Nm}(\rho - (-1)) = -f_a(-1) = -(-1 + (a-1) + a - 1) = -(2a - 3) = 3 - 2a. \quad \square$$

5.2. **Exact computation of n_K and μ_K.** All values below were computed by Voronoi's neighbour algorithm run in exact arithmetic. We record the ingredients, since they are what make the numbers trustworthy.

Lemma 5.2. *Let* K *be totally real of degree* 3 *and* $\mathfrak{o}$ *an order. Every perfect form with* $m_{\mathfrak{o}}(a) = 1$ *is of the shape* $a = (\sigma_1(\alpha), \sigma_2(\alpha), \sigma_3(\alpha))$ *for a totally positive* $\alpha \in K$, *i.e.* $a[x] = \mathrm{Tr}_{K/\mathbb{Q}}(\alpha x^2)$*. Two such forms* α, α' *are* $\Gamma_{\mathfrak{o}}$*–equivalent iff* $\alpha'/\alpha = u^2$ *with* $u \in \mathfrak{o}^\times$.

Proof. Every $a \in \mathbb{R}^3$ is $\sigma(\alpha)$ for a unique $\alpha \in K \otimes \mathbb{R}$, and then $\langle a, q(x) \rangle = \mathrm{Tr}(\alpha x^2)$. If a is perfect with minimal vectors x_1, x_2, x_3 whose squares are linearly independent, α is the unique solution of $\mathrm{Tr}(\alpha x_i^2) = 1$ ($i = 1, 2, 3$). In a $\mathbb{Q}$–basis (b_k) of K this is a linear system with rational coefficients $\mathrm{Tr}(b_k x_i^2)$, so $\alpha \in K$. The second assertion is Lemma 3.3(b). □

Thus every vertex of $K_1(m_\mathfrak{o})$ is represented exactly. The algorithm starts from the trace form $\alpha = 1/3$ and walks inside faces until the minimal vectors have full rank. At a perfect form it lists the edges of the dual polygon $\mathrm{conv}\{q(x) : x \in S_\mathfrak{o}(a)\}$ and walks along each edge to the neighbouring perfect form. It stops when every neighbour is equivalent to a form already found. Three points are essential.

(i) *Minimal vectors are computed exactly.* The Gram matrix of $x \mapsto \mathrm{Tr}(\alpha x^2)$ on a $\mathbb{Z}$–basis is rational. After an LLL change of basis we enumerate $\{x : \mathrm{Tr}(\alpha x^2) \le B\}$ from the exact rational Gauss decomposition. Floating point is used only to widen loop ranges, never to decide membership.

(ii) *Equivalences are tested exactly.* $\alpha \sim \alpha'$ is decided by checking that α'/α is integral, of norm 1, and a square of an element of $\mathfrak{o}$; $x \sim x'$ by checking that x'/x is integral of norm ± 1. No unit group computed by `bnfinit` (which is conditional on GRH) enters any decision; fundamental units are used only to keep heights small.

(iii) *Two certificates.* The output must satisfy $V - E + F = 0$ (Theorem 4.3). Moreover the units u produced by the equivalences found along edges must generate the unit group, by the next proposition.

Proposition 5.3. *Let $r_1, \dots, r_{n_K}$ be representatives of the perfect forms modulo $\Gamma_\mathfrak{o}$, and let S be the set of $u \in \mathfrak{o}^\times/\{\pm 1\}$ such that some neighbour of some r_i equals $r_j \cdot u$. Then S generates $\mathfrak{o}^\times/\{\pm 1\}$.*

Proof. The group $G = \mathfrak{o}^\times/\{\pm 1\}$ acts freely on the 1–skeleton $X^{(1)}$ of $\partial K_1(m_\mathfrak{o})$, which is connected because $\partial K_1(m_\mathfrak{o}) \cong \mathbb{R}^2$ carries a locally finite polyhedral structure. Let $g \in G$ and choose an edge path $r_1 = v_0, v_1, \dots, v_N = g \cdot r_1$. Write $v_k = g_k \cdot r_{i_k}$ with $g_0 = 1$. Since v_{k+1} is a neighbour of v_k, the form $g_k^{-1} v_{k+1}$ is a neighbour of r_{i_k} and equals $(g_k^{-1} g_{k+1}) \cdot r_{i_{k+1}}$, so $g_k^{-1} g_{k+1} \in S$. By induction $g = g_N \in \langle S \rangle$. □

In every computation reported below, the elements of S lie in the subgroup generated by the fundamental units returned by PARI's `bnfinit` and generate it with index 1. By Proposition 5.3 this proves that those units are fundamental, so every regulator quoted below is certified without GRH.

5.3. **The simplest families.**

Computational Observation 5.4 (Ennola and Shanks families)**.** For Ennola's fields with $2 \le a \le 12$ the exact computation gives

a	2	3	4	5	6	7	8	9	10	11	12
disc	49	257	697	1489	2777	4729	7537	11417	16609	23377	32009
Reg_K	0.525	1.975	2.712	3.361	3.949	4.491	4.995	5.467	5.912	6.334	6.734
(μ_K, n_K)	(1, 2)	(2, 4)	(1, 2)	(1, 2)	(1, 2)	(1, 2)	(1, 2)	(1, 2)	(1, 2)	(1, 2)	(1, 2)

At $a = 3$ the second extreme square is $1 + \rho$, of norm $3 - 2a = -3$ (Lemma 5.1). For Shanks' simplest cubic fields $x^3 - ax^2 - (a+3)x - 1$ with $-1 \le a \le 11$ one finds $(\mu_K, n_K) = (1, 2)$ throughout. In all these cases the torus carries the one–hexagon structure $V = 2$, $E = 3$, $F = 1$ of Corollary 4.5. We have not proved that $\mu_K = 1$ for all $a \ge 4$; by Remark 6.6 the lower bound of Theorem 6.5 does not exclude it.

Computational Observation 5.5. For the field of discriminant 404 ($x^3 - 5x^2 + x + 1$) one has $(\mu_K, n_K) = (4, 7)$. The degrees of the seven vertices are $3, 3, 3, 3, 3, 3, 4$, in accordance with the defect formula (2).

5.4. **All fields of discriminant at most** 20000**.** By Hunter's theorem (see [5]), every totally real cubic field of discriminant $D \le 20000$ is generated by a root of $x^3 + ax^2 + bx + c$ with $a \in \{0, -1\}$ and $a^2 - 2b \le \frac{1}{3} + (4/3)^{1/2}(D/3)^{1/2}$. Hence $b \ge -47$, $|c| \le 178$, and the polynomial discriminant is at most $(a^2 - 2b)^3 < 9 \cdot 10^5$. Enumerating this box and reducing with `polredabs` yields 832 fields. This is therefore the complete list.

Computational Observation 5.6 (Census)**.** For all 832 totally real cubic fields with $D \le 20000$:

(a) both certificates of §5.2 hold, and the inequalities of Theorem 6.5 hold;
(b) $\mu_K = 1$ for exactly 27 fields; the maximum is $n_K = 152$, $\mu_K = 77$, attained at $D = 16689$ ($x^3 - x^2 - 26x + 24$, $\mathrm{Reg}_K = 110.63$);
(c) 68% of the fields are simplicial; the maximal vertex degree is 6, attained in 10 fields;
(d) among the 686 fields with $h_K = 1$ and $\mathrm{Reg}_K > 10$, the ratio μ_K/Reg_K has median 0.62 and lies in $[0.24, 0.95]$.

Computational Observation 5.7 (Census, $D \le 10^5$)**.** For all 4804 totally real cubic fields with $D \le 10^5$ (the complete list, from PARI's `nflist`, identical field by field with the LMFDB), both certificates hold and:

(a) $n_K \notin \{1, 3\}$; the smallest values are $2, 4, 5, 6, \ldots$, and $\mu_K = 1$ for exactly 55 fields. The maximum is $n_K = 500$, $\mu_K = 253$, at $D = 82657$ ($x^3 - 49x - 72$, $\mathrm{Reg}_K = 396.0$);
(b) the inequalities of Theorem 6.5 hold, and $\frac{4}{9}\,\ell_{\mathfrak{o}} \le n_K \le 2\ell_{\mathfrak{o}}$; the median of $n_K/\ell_{\mathfrak{o}}$ lies between 1.158 and 1.185 in each discriminant window of width $2 \cdot 10^4$;
(c) among the 4112 fields with $h_K = 1$ and $\mathrm{Reg}_K > 10$, the median of μ_K/Reg_K is 0.56 (range $[0.13, 0.95]$); by window it drifts down, $0.62, 0.58, 0.56, 0.56, 0.55$, as does the share of simplicial fields, from 68% to 60%;
(d) a vertex of degree 6 occurs in 38 fields, and Leibak's bound is attained in 306 (§8).

Larger regulators are equally accessible; the run time grows roughly linearly in n_K:

polynomial	D_K	Reg_K	μ_K	n_K	μ_K/Reg_K
$x^3 - x^2 - 221x + 29$	676829	262.8	136	272	0.52
$x^3 - x^2 - 244x + 29$	58271449	1186.7	546	1092	0.46
$x^3 - x^2 - 382x + 7$	223164633	3741.8	1767	3534	0.47

In these three fields every perfect form is simplicial.

Remark 5.8 (Correction to an earlier draft)**.** An earlier version of this note computed μ_K by enumerating a box of lattice points and testing extremality by linear programming. Those values are superseded. Extremality relative to a finite subset is weaker than extremality in C, so that method can over–count as well as under–count; the earlier claim that it only under–counts was false. In addition, for $x^3 - x^2 - 10x + 1$, whose polynomial discriminant is $3^2 \cdot 473$, the earlier computation evidently used $\mathbb{Z}[\theta]$ instead of $\mathcal{O}_K$. The earlier values $(\mu_K, n_K) = (6, 12)$ at $D = 473$ and $\mu_K = 3$ at $D = 6856$ are wrong; the correct values are $(2, 4)$ and $(23, 46)$. The conclusion drawn from them, that μ_K is not controlled by the regulator, is also wrong: Theorem 6.5 shows that it is, up to a factor $(\log D_K)^2$.

6. The regulator sandwich

Throughout this section K is a totally real cubic field and $\mathfrak{o}$ an order of K, with discriminant $D = D_{\mathfrak{o}} > 0$ and regulator $\mathrm{Reg} = \mathrm{Reg}_{\mathfrak{o}}$ of $\mathfrak{o}^\times$. For $x \in K^\times$ put $\ell(x) = (\log|\sigma_i(x)|)_{i=1}^3$. Let

$H_0 = \{p \in \mathbb{R}^3 : \sum p_i = 0\}$, let π be the orthogonal projection onto H_0, and let $\Lambda = \ell(\mathfrak{o}^\times) \subset H_0$, a lattice of covolume $\sqrt{3}\,\mathrm{Reg}$ for the Euclidean metric. On H_0 we use the *range norm*

$$r(p) = \max_i p_i - \min_i p_i.$$

It is a norm on H_0 and is invariant under adding multiples of $(1,1,1)$, so $r(\pi(v)) = r(v)$ for all $v \in \mathbb{R}^3$.

Lemma 6.1. *The ball $\{p \in H_0 : r(p) \le \rho\}$ has area $\sqrt{3}\,\rho^2$.*

Proof. It is the orthogonal projection of the cube $[0,\rho]^3$ onto H_0. Indeed $p \in H_0$ with $r(p) \le \rho$ is the projection of $p - (\min_i p_i)(1,1,1) \in [0,\rho]^3$, and conversely every point of the cube has range at most ρ. The projection of a unit cube onto a plane with unit normal ν has area $\sum |\nu_i|$, which is $\sqrt{3}$ here. □

Definition 6.2. A nonzero $x \in \mathfrak{o}$ is a *relative minimum* of $\mathfrak{o}$ if there is no $y \in \mathfrak{o} \setminus \{0, \pm x\}$ with $|\sigma_i(y)| \le |\sigma_i(x)|$ for $i = 1,2,3$. Let $\ell_{\mathfrak{o}}$ be the number of $\mathfrak{o}^\times$–classes of relative minima.

Relative minima of the lattice $\iota(\mathfrak{o})$ are the objects enumerated by Voronoi's algorithm for cubic fields [24, 7]; see Buchmann [2, 3] for its generalisations and period lengths. The following lemma is the precise link between the squared sail and that genuinely linear object. The map q is not additive, but it preserves coordinatewise domination.

Lemma 6.3. *If $q(x)$ is a vertex of C, then x is a relative minimum of $\mathfrak{o}$. Hence $\mu_K \le \ell_{\mathfrak{o}}$.*

Proof. Suppose $y \neq 0, \pm x$ satisfies $|\sigma_i(y)| \le |\sigma_i(x)|$ for all i. Then $v := q(x) - q(y) \ge 0$. Also $v \neq 0$, since otherwise $y/x \in K$ would have all conjugates ± 1 and would therefore equal ± 1. Then $q(x) = \frac{1}{2}q(y) + \frac{1}{2}\big(q(y) + 2v\big)$ is the midpoint of two distinct points of C, so it is not a vertex. □

Lemma 6.4 (Separation)**.** *Let $x \neq \pm y$ be relative minima such that $\sigma_i(x)\sigma_i(y)$ has the same sign for $i = 1,2,3$. Then $r\big(\ell(x) - \ell(y)\big) \ge 2\log 2$.*

Proof. Replacing y by $-y$ we may assume $\sigma_i(x)$ and $\sigma_i(y)$ have the same sign for every i. Suppose $|\sigma_i(y)| < 2|\sigma_i(x)|$ for all i, and put $z = x - y$. Then $z \neq 0$, and $z \neq -x$ because $y \neq 2x$. For each i we have $|\sigma_i(z)| = \big||\sigma_i(x)| - |\sigma_i(y)|\big| < |\sigma_i(x)|$: if $|\sigma_i(y)| \le |\sigma_i(x)|$ this uses $\sigma_i(y) \neq 0$, and otherwise it uses $|\sigma_i(y)| < 2|\sigma_i(x)|$. This contradicts the minimality of x. Hence $\max_i(\ell_i(y) - \ell_i(x)) \ge \log 2$. By symmetry $\min_i(\ell_i(y) - \ell_i(x)) \le -\log 2$. □

Theorem 6.5 (Regulator sandwich)**.** *Let K be a totally real cubic field and $\mathfrak{o}$ an order in K. Then*

$$\boxed{\frac{4\,\mathrm{Reg}}{(\log 2D)^2} \;\le\; \mu_K \;\le\; \ell_{\mathfrak{o}} \;\le\; \frac{4\,\mathrm{Reg}}{(\log 2)^2}} \qquad \text{and} \qquad \frac{2\,\mathrm{Reg}}{(\log 2D)^2} \;\le\; n_K \;\le\; \frac{8\,\mathrm{Reg}}{(\log 2)^2}.$$

Proof. Upper bound. Let $U^+ \subseteq \mathfrak{o}^\times$ be the subgroup of units whose three conjugates have the same sign, and let $\iota = [\mathfrak{o}^\times : U^+]$. Since $\pm 1 \in U^+$, the lattice $\ell(U^+)$ has index ι in Λ, and hence covolume $\iota\sqrt{3}\,\mathrm{Reg}$. The sign vectors of nonzero elements, modulo ± 1, fall into four classes c, and U^+ preserves each class. Fix c and let $P_c \subset H_0$ be the set of points $\pi\ell(x)$, where x runs over relative minima of class c. By Lemma 6.4, distinct points of P_c are at range distance at least $2\log 2$. Since P_c is $\ell(U^+)$–invariant, the range balls of radius $\log 2$ about its points are pairwise disjoint and embed in the torus $H_0/\ell(U^+)$. By Lemma 6.1 the number N_c of U^+–orbits in class c satisfies $N_c\sqrt{3}(\log 2)^2 \le \iota\sqrt{3}\,\mathrm{Reg}$. The group $\mathfrak{o}^\times/\{\pm 1\}$ acts freely on relative minima modulo sign, so each $\mathfrak{o}^\times$–orbit consists of exactly ι orbits of U^+. Therefore $\iota\,\ell_{\mathfrak{o}} = \sum_c N_c \le 4\iota\,\mathrm{Reg}\,/(\log 2)^2$. Together with Lemma 6.3 this gives the upper bounds for μ_K and $\ell_{\mathfrak{o}}$.

Lower bound. Let $a \in \mathbb{R}^3_{>0}$ with $a_1a_2a_3 = 1$. On a $\mathbb{Z}$–basis of $\mathfrak{o}$, the form $x \mapsto \langle a, q(x)\rangle$ has Gram determinant $a_1a_2a_3\, D = D$. Since $\gamma_3^3 = 2$, Hermite's inequality gives $m_{\mathfrak{o}}(a) \le B := 2^{1/3}D^{1/3}$. By Proposition 3.4 the face of C exposed by a is the polygon $\operatorname{conv}\{q(x) : x \in S_{\mathfrak{o}}(a)\}$, and its vertices are extreme squares. So there is an extreme square $q(x)$ with $\langle a, q(x)\rangle = m_{\mathfrak{o}}(a)$. Each of the three terms $t_i = a_i\sigma_i(x)^2$ is therefore at most B, while $t_1t_2t_3 = \operatorname{Nm}(x)^2 \ge 1$. Hence each $t_i \ge B^{-2}$, and

$$r\big(2\ell(x) + \log a\big) = \max_i \log t_i - \min_i \log t_i \le 3\log B = \log 2D.$$

As a ranges over the forms with $a_1a_2a_3 = 1$, the vector $-\frac{1}{2}\log a$ ranges over all of H_0. Thus every point of H_0 lies within range distance $\frac{1}{2}\log 2D$ of $\pi\ell(x)$ for some extreme square $q(x)$. These points form finitely many Λ–orbits, at most μ_K of them. Their range balls of radius $\frac{1}{2}\log 2D$ therefore cover the torus H_0/Λ, of area $\sqrt{3}\operatorname{Reg}$, and Lemma 6.1 gives $\mu_K \cdot \frac{\sqrt{3}}{4}(\log 2D)^2 \ge \sqrt{3}\operatorname{Reg}$.

Finally, $\frac{1}{2}\mu_K \le n_K \le 2\mu_K$ by Theorem 4.3. □

Remark 6.6 (Sharpness)**.** The lower bound is sharp up to a constant factor. In Ennola's and Shanks' families $\operatorname{Reg}_K \sim (\log a)^2$ and $D_K \asymp a^4$, so $4\operatorname{Reg}_K/(\log 2D_K)^2 \to \frac{1}{4}$. The computed values are 0.22 at $a = 12$ in both families (Observation 5.4), while $\mu_K = 1$. For such families the theorem only guarantees $\mu_K \ge 1$, so it is consistent with, but does not explain, the observed floor $\mu_K = 1$. The upper bound is of the right order in the generic case: in the census, $\mu_K/\operatorname{Reg}_K$ is typically about 0.6 (Observation 5.6). The constant $4/(\log 2)^2 \approx 8.3$ is not optimal.

Remark 6.7. Replacing Hermite's inequality by the additive Hermite–Humbert constant $\gamma_K = \sup_a m(a)/(a_1a_2a_3)^{1/3}$ of Leibak [15] in the proof gives $\mu_K \ge 4\operatorname{Reg}_K/(3\log\gamma_K)^2$. In the other direction, Porter and Mendelsohn [21] bound the number of classes of perfect unary forms over any totally real K of degree n by $|\Delta_K|^{-1/2}\gamma_K^n \exp(2n\rho_\infty(\Lambda_K))$, where ρ_∞ is the ℓ^∞ covering radius of the log–unit lattice. For cubic fields the bound $n_K \le 8\operatorname{Reg}_K/(\log 2)^2$ is much stronger, being linear in Reg_K instead of exponential in the covering radius.

Corollary 6.8. (a) *Over totally real cubic fields,* $\sup n_K = \infty$ *as soon as* $\sup \operatorname{Reg}_K/(\log D_K)^2 = \infty$.

(b) *If K ranges over totally real cubic fields with $h_K \le D_K^{1/2-\delta}$, then $n_K \ge D_K^{\delta - o(1)}$.*

(c) *For a fixed K and for infinitely many primes p, the order $\mathfrak{o}_p = \mathbb{Z} + p\mathcal{O}_K$ satisfies $\mu_{\mathfrak{o}_p} \gg_K p^{1/3}/(\log p)^2$. In particular* $\sup_p n_{\mathfrak{o}_p} = \infty$ *unconditionally.*

Proof. (a) is immediate from Theorem 6.5. (b) follows from the Brauer–Siegel theorem [1], $h_K \operatorname{Reg}_K = D_K^{1/2+o(1)}$. For (c), let ε be a fundamental unit of $\mathcal{O}_K$. Since $\ell(\mathfrak{o}_p^\times)$ has index $[\mathcal{O}_K^\times : \mathfrak{o}_p^\times]$ in $\ell(\mathcal{O}_K^\times)$, we have $\operatorname{Reg}_{\mathfrak{o}_p} = [\mathcal{O}_K^\times : \mathfrak{o}_p^\times]\operatorname{Reg}_K$, and $D_{\mathfrak{o}_p} = p^4 D_K$ because $[\mathcal{O}_K : \mathfrak{o}_p] = p^2$. The index is at least the least $k \ge 1$ with $\varepsilon^k \in \mathfrak{o}_p$. For each k, the element $\varepsilon^k \notin \mathbb{Z}$ has a nonzero coordinate c_k outside 1 in a $\mathbb{Z}$–basis $1, \omega_1, \omega_2$ of $\mathcal{O}_K$, and $\varepsilon^k \in \mathfrak{o}_p$ forces $p \mid c_k$. As $\log|c_k| \ll_K k$, at most $O_K(k)$ primes divide c_k. Hence at most $O_K(Y^2)$ primes have index $\le Y$. Taking $Y = X^{1/3}$, all but $O(X^{2/3})$ of the $\gg X/\log X$ primes $p \le X$ have index $> p^{1/3}$, and Theorem 6.5 applies. □

Cohen and Martinet [6] predict that a positive proportion of totally real cubic fields have $h_K = 1$. Combined with Corollary 6.8(b), this predicts $n_K = D_K^{1/2+o(1)}$ for a positive proportion of fields, which is consistent with Observation 5.6. In this sense the "opposite extreme" to the simplest cubic fields is the generic field. We do not know an unconditional proof that $\operatorname{Reg}_K/(\log D_K)^2$ is unbounded over maximal orders; see §10.

Remark 6.9. Unit–reducible fields in the sense of Leibak–Porter–Ling [18] have $\mu_K = 1$, since every extreme square is a minimal vector. Theorem 6.5 therefore shows that a unit–reducible totally real cubic field satisfies $\mathrm{Reg}_K \le \frac{1}{4}(\log 2D_K)^2$.

7. Relative minima, Klein sails and the Minkowski–Voronoi complex

Throughout this section K is a totally real cubic field, $\mathfrak{o}$ an order of K, and $\Lambda_\mathfrak{o} = \iota(\mathfrak{o}) \subset \mathbb{R}^3$. Lemma 6.3 places the extreme squares among the relative minima of $\Lambda_\mathfrak{o}$. We now compare the relative minima with the classical *Klein sails* of $\Lambda_\mathfrak{o}$, and record the cell structure of the relative minima themselves.

7.1. **Klein sails.** For $\theta \in \{\pm 1\}^3$ let $C_\theta = \{x \in \mathbb{R}^3 : \theta_i x_i > 0\}$ be the open orthant and $\mathcal{K}_\theta = \mathrm{conv}(\Lambda_\mathfrak{o} \cap C_\theta)$ the *Klein polyhedron*; its boundary $\partial\mathcal{K}_\theta$ is the *sail* of θ. The unit group acts on the union of the sails: $u \in \mathfrak{o}^\times$ maps $\mathcal{K}_\theta$ onto $\mathcal{K}_{\theta'}$, $\theta'_i = \theta_i \operatorname{sign}\sigma_i(u)$. We write $V_\mathfrak{o}$, $E_\mathfrak{o}$, $F_\mathfrak{o}$ for the numbers of $\mathfrak{o}^\times$–classes of vertices, edges and facets (compact faces) of all the sails. No lattice point lies on a coordinate plane, since $\mathrm{Nm}(x) \neq 0$ for $x \neq 0$.

Proposition 7.1. *Modulo $\mathfrak{o}^\times$ the union of the sails is a finite disjoint union of 2–tori, with a cell structure whose face numbers satisfy*

$$V_\mathfrak{o} - E_\mathfrak{o} + F_\mathfrak{o} = 0, \qquad \tfrac{1}{2}V_\mathfrak{o} \le F_\mathfrak{o} \le 2V_\mathfrak{o}.$$

Proof. Fix θ. Every point of $\mathcal{K}_\theta$ has $\prod_i |x_i| \ge 1$, because $\prod_i |x_i|^{1/3}$ is concave on C_θ and is at least 1 on $\Lambda_\mathfrak{o} \cap C_\theta$. The set $\mathcal{K}_\theta$ is a generalised polyhedron with recession cone $\overline{C_\theta}$ and finitely many faces modulo the totally positive units (see [13]). Hence every open ray from 0 in C_θ meets $\partial\mathcal{K}_\theta$ exactly once, and radial projection identifies $\partial\mathcal{K}_\theta$ with $C_\theta/\mathbb{R}_{>0} \cong \mathbb{R}^2$. In the logarithmic chart of Lemma 4.1 the stabiliser of θ in $\mathfrak{o}^\times/\{\pm 1\}$, the totally positive units, acts by translations through a lattice of finite index in $L(\mathfrak{o}^\times)$. This action is free, properly discontinuous and cocompact, and the faces of the sail are bounded, so as in the proof of Theorem 4.3 the quotient is a torus with an honest finite cell structure. Each orbit of orthants under $\mathfrak{o}^\times$ contributes one such torus. Every vertex of a sail has degree at least 3 and every facet has at least 3 sides. The inequalities follow from Euler's formula exactly as (4) gives $\frac{1}{2}\mu_K \le n_K \le 2\mu_K$. □

The next two results are due to Bykovskii and German; we state them in our setting. Bykovskii and German use the term "local minimum" and the closed orthant, respectively. Both coincide with our definitions for $\Lambda_\mathfrak{o}$: $|\sigma_i(y)| = |\sigma_i(x)|$ forces $y = \pm x$, and no lattice point lies on a coordinate plane.

Theorem 7.2 (Bykovskii [4]; German [9])**.** *Let K be totally real cubic and $\mathfrak{o}$ an order of K.*

(a) *Every vertex of every sail is a relative minimum of $\mathfrak{o}$* [4, Theorem].
(b) *Every relative minimum lies on the sail of its orthant* [9, Thm. 1].
(c) *A relative minimum x that is not a sail vertex is a* German point*: there are a triangular facet* $\mathrm{conv}(u_1, u_2, u_3)$ *of its sail and an integer $k \ge 1$ with*

$$x = \frac{k}{2k+1}(u_1 + u_2) + \frac{1}{2k+1}u_3,$$

and the lattice points of the facet are exactly u_1, u_2, u_3 and $u_3 + \frac{m}{k}(x - u_3)$, $1 \le m \le k$. Conversely every such point is a relative minimum [9, Cor. 2, Thm. 2, Cor. 3].

The hypotheses of [9, Thm. 1, Cor. 3] are satisfied. German's exceptional lattices have nonzero points on the coordinate axes, and $\Lambda_\mathfrak{o}$ has no nonzero point on any coordinate plane. In dimension 2 the analogue is classical: for a real quadratic field the sail vertices modulo units are exactly the relative minima [4], i.e. the ℓ_d convergents of one period.

Corollary 7.3. *Every extreme square is a sail vertex or a German point.*

Proof. Combine Lemma 6.3 with Theorem 7.2. □

Theorem 7.4. *Let $G_{\mathfrak{o}}$ be the number of $\mathfrak{o}^\times$–classes of German points. Then $\ell_{\mathfrak{o}} = V_{\mathfrak{o}} + G_{\mathfrak{o}}$, each facet contains at most one German point, and*

$$\tfrac{1}{3}\,\ell_{\mathfrak{o}} \;\le\; V_{\mathfrak{o}} \;\le\; \ell_{\mathfrak{o}}, \qquad \frac{4\,\mathrm{Reg}}{3(\log 2D)^2} \;\le\; V_{\mathfrak{o}} \;\le\; \frac{4\,\mathrm{Reg}}{(\log 2)^2}, \qquad \frac{2\,\mathrm{Reg}}{3(\log 2D)^2} \;\le\; F_{\mathfrak{o}} \;\le\; \frac{8\,\mathrm{Reg}}{(\log 2)^2}.$$

Moreover $n_K \le 2\mu_K \le 2\ell_{\mathfrak{o}} \le 6V_{\mathfrak{o}}$.

Proof. By Theorem 7.2 the relative minima are the sail vertices together with the German points, and distinct classes of vertices are distinct classes of relative minima; so $\ell_{\mathfrak{o}} = V_{\mathfrak{o}} + G_{\mathfrak{o}}$ and $V_{\mathfrak{o}} \le \ell_{\mathfrak{o}}$. A German point determines k as the number of lattice points of its facet minus 3. For $k = 1$ it is the barycentre. For $k \ge 2$ the interior lattice points of the facet span a segment of the median through a single vertex u_3, and the point is the end of that segment opposite to u_3. So a facet carries at most one German point, and $G_{\mathfrak{o}} \le F_{\mathfrak{o}} \le 2V_{\mathfrak{o}}$ by Proposition 7.1. Hence $\ell_{\mathfrak{o}} \le 3V_{\mathfrak{o}}$. The bounds in terms of Reg follow from Theorem 6.5, which gives $4\,\mathrm{Reg}\,/(\log 2D)^2 \le \mu_K \le \ell_{\mathfrak{o}} \le 4\,\mathrm{Reg}\,/(\log 2)^2$, together with $\frac{1}{2}V_{\mathfrak{o}} \le F_{\mathfrak{o}} \le 2V_{\mathfrak{o}}$. The last chain is Theorem 4.3, Lemma 6.3 and $\ell_{\mathfrak{o}} \le 3V_{\mathfrak{o}}$. □

Remark 7.5. For sails, the volume of the region under the sail gives a bound of order $\sqrt{D}$ Reg on the number of facets modulo totally positive units. That region lies in $\{\prod_i |x_i| \le 2D/27\}$, by Hermite's inequality for the facet functionals, and the cone over each facet has volume at least $\sqrt{D}/6$. Theorem 7.4 replaces this, for cubic fields, by a bound linear in Reg with an absolute constant, and supplies the matching lower bound up to $(\log D)^2$. The bridge in both directions is the set of relative minima: it contains the extreme squares and the sail vertices, and it is separated in logarithmic space (Lemma 6.4).

Lemma 7.6. *A totally positive relative minimum x of $\mathfrak{o}$ is indecomposable, i.e. not a sum of two totally positive elements of $\mathfrak{o}$. In particular every totally positive extreme square, sail vertex and German point is indecomposable.*

Proof. If $x = y + z$ with $y, z \gg 0$, then $0 < \sigma_i(y) < \sigma_i(x)$ for $i = 1, 2, 3$, so $y \ne 0, \pm x$ is dominated by x, contradicting minimality. □

7.2. **The Minkowski–Voronoi complex.** Following Karpenkov–Ustinov [14, §2], call relative minima x, y *adjacent* if the box $\{z : |\sigma_i(z)| \le \max(|\sigma_i(x)|, |\sigma_i(y)|)\}$ contains no lattice point other than $0, \pm x, \pm y$, and call $\{x, y, z\}$ a *minimal triple* if the box of x, y, z contains no other nonzero lattice point up to sign and each of x, y, z attains one of the three coordinate maxima. The Minkowski–Voronoi complex has the relative minima as 2–cells, the adjacent pairs as edges and the minimal triples as vertices [14, Def. 2.5].

Proposition 7.7. *Modulo $\mathfrak{o}^\times$ the Minkowski–Voronoi complex of $\Lambda_{\mathfrak{o}}$ is a 2–torus with f–vector*

$$(V, E, F) = (2\ell_{\mathfrak{o}},\ 3\ell_{\mathfrak{o}},\ \ell_{\mathfrak{o}}).$$

Proof. Let $S = \{(|\sigma_i(x)|)_i : 0 \ne x \in \mathfrak{o}\} + \mathbb{R}^3_{\ge 0}$. The complex is the cell structure of the staircase ∂S [14, Def. 2.8, Prop. 2.11]. Its outer corners are the relative minima and its inner corners the minimal triples. These descriptions are local, and KU state them for finite sets in general position. The set $\{(|\sigma_i(x)|)_i\}$ is in general position because $|\sigma_i(y)| = |\sigma_i(x)|$ forces $y = \pm x$. The set S is closed under scaling up, every point of S has coordinate product at least 1, and every open ray in $\mathbb{R}^3_{>0}$ meets S by Minkowski's convex body theorem. So radial projection identifies ∂S with $\mathbb{R}^2$, and $\mathfrak{o}^\times$ acts on it by the translations $L(u)$. As in Theorem 4.3 the quotient is a torus, and $V - E + F = 0$. In general position the three 2–subsets of a minimal triple are adjacent

pairs, the three edges at that vertex [14, §2.3.1]; so every vertex has degree 3 and $2E = 3V$. Hence $V = 2F$ and $E = 3F$, with $F = \ell_{\mathfrak{o}}$. □

Lemma 7.8. *A minimal triple $\{x, y, z\}$ of $\Lambda_{\mathfrak{o}}$ is either a $\mathbb{Z}$–basis of $\mathfrak{o}$ or, after changing the signs of its members, satisfies $z = x + y$.*

Proof. Let $m_i = \max(|x_i|, |y_i|, |z_i|)$. Then x, y, z attain different maxima, say $|x_1| = m_1$, $|y_2| = m_2$, $|z_3| = m_3$, and all other entries are strictly smaller in absolute value. The box $B = \prod_i [-m_i, m_i]$ contains no lattice point besides $0, \pm x, \pm y, \pm z$, and these lie on its boundary. Suppose first that x, y, z are independent and let $L = \mathbb{Z}x + \mathbb{Z}y + \mathbb{Z}z$. By Minkowski's theorem $8m_1m_2m_3 \le 8\,\mathrm{covol}(\Lambda_{\mathfrak{o}})$, and $|\det(x, y, z)| \le 4m_1m_2m_3$, because a determinant with entries bounded by 1 in absolute value is at most 4. Hence $[\mathfrak{o} : L] \le 4$. An element of order 2 of $\mathfrak{o}/L$ is represented by $(ax + by + cz)/2$ with $a, b, c \in \{0, 1\}$, not all 0, and arbitrary signs. With one or two nonzero terms this point lies in the interior of B. With three, coordinate i of $x + sy + tz$ reaches $2m_i$ in absolute value only if its three terms have the same sign, which happens for one choice of (s, t) at most. So one of the four choices gives an interior point. An element of order 3 is represented by $(ax + by + cz)/3$ with $a, b, c \in \{-1, 0, 1\}$, again an interior point. So $[\mathfrak{o} : L]$ is prime to 6 and at most 4, i.e. $L = \mathfrak{o}$.

Suppose now that $z = px + qy$ with $p, q \in \mathbb{Q}$. The pair x, y is adjacent, and the same argument in the plane shows that it is a basis of $\mathfrak{o} \cap (\mathbb{Q}x + \mathbb{Q}y)$; so $p, q \in \mathbb{Z}$, and after changing signs $p, q \ge 0$. If $q = 0$ or $p = 0$, then z is dominated by x or y. If $p, q \ge 1$ and $(p, q) \ne (1, 1)$, then

$$x + y = \tfrac{q-1}{p+q-1}\, x + \tfrac{p-1}{p+q-1}\, y + \tfrac{1}{p+q-1}\, z$$

is a convex combination with at least two positive weights, hence an interior point of B. So $z = x + y$. □

In the 832 fields with $D_K \le 20000$, all 50,176 classes of minimal triples are of the two kinds, as the lemma says, and 12,903 of them (25.7%) are degenerate. In every field at most half are degenerate, and only $D_K = 81$ has none. Each of the 26 fields with $\ell_{\mathfrak{o}} = 1$ and $D_K \le 10^6$ has one basis class and one degenerate class, both made of units. $D_K = 81$ ($\ell = 2$) has four basis classes.

Computational Observation 7.9. These are certified computations.

(a) For all 832 fields with $D \le 20000$, a direct count of the minimal triples and adjacent pairs gives $(V, E, F) = (2\ell, 3\ell, \ell)$. The relative minima that are not sail vertices are German points with $k = 1, \dots, 7$. Their facets have integer distance 1 from the origin, except for 17 barycentres on facets at distance 3.

(b) For all 4804 fields with $D \le 10^5$, $\mu_K \le V_{\mathfrak{o}} \le \ell_{\mathfrak{o}}$ and $V_{\mathfrak{o}} \ge \frac{1}{2}\ell_{\mathfrak{o}}$. Of the 202,071 classes of extreme squares, 201,948 are sail vertices; the other 123 are German points, 121 with $k = 1$ and 2 with $k = 2$ ($D = 61108$, 75425), all on facets at distance 1.

8. Equality in Leibak's bound

Let K be totally real of degree m, $\mathcal{O}_K$ its ring of integers, $\mathfrak{d}$ its different and

$$\gamma_K = \sup_a \frac{m_{\mathcal{O}_K}(a)}{(a_1 \cdots a_m)^{1/m}}$$

its additive Hermite–Humbert constant (Remark 6.7). The form $x \mapsto \langle a, q(x)\rangle$ on $\mathcal{O}_K \cong \mathbb{Z}^m$ has determinant $a_1 \cdots a_m D_K$, so Hermite's inequality gives Leibak's bound $\gamma_K^m \le \gamma_m^m D_K$ [15, Thm. 6], quoted as [16, Thm. 2.2]. Leibak notes that it is attained for $\mathbb{Q}(\sqrt{3})$ [16, after Thm. 2.2]. On a face of $\partial K_1(m_{\mathcal{O}_K})$ one has $m_{\mathcal{O}_K}(a) = 1$, and $a \mapsto (a_1 \cdots a_m)^{1/m}$ is concave. So the supremum, which is a maximum by compactness modulo units, is attained at a vertex, i.e. a

perfect form $a = \iota(\alpha)$. Then $\gamma_K^m = 1/\min \mathrm{Nm}(\alpha)$ over the perfect forms of minimum 1. Let $\Lambda_2 = A_2$ and $\Lambda_3 = A_3$, scaled to be even with minimum 2, so that $\det \Lambda_2 = 3$ and $\det \Lambda_3 = 4$.

Theorem 8.1. *Let $m \in \{2, 3\}$. The following are equivalent:*

(i) $\gamma_K^m = \gamma_m^m D_K$;
(ii) *there is a totally positive $\beta \in \mathfrak{d}^{-1}$ with $\mathrm{Nm}(\beta\mathfrak{d}) = \det \Lambda_m$ and $\mathrm{Tr}(\beta x^2) \in 2\mathbb{Z}$ for all $x \in \mathcal{O}_K$.*

In that case 2 ramifies in K and $\mathcal{O}_K/\beta\mathfrak{d} \cong \Lambda_m^{\#}/\Lambda_m$. For $m = 2$ the ideal $\beta\mathfrak{d}$ is a prime of norm 3. For $m = 3$ one has $2\mathcal{O}_K = \mathfrak{p}\mathfrak{q}^2$ and $\beta\mathfrak{d} = \mathfrak{p}^2$, with $\mathfrak{p}, \mathfrak{q}$ of degree 1.

Proof. (i)⇒(ii). Equality means that the lattice $L_\alpha = (\mathcal{O}_K, \mathrm{Tr}(\alpha xy))$ of some perfect α attains γ_m. The critical lattice is unique up to similarity: the hexagonal lattice for $m = 2$ and the face–centred cubic lattice for $m = 3$. So $L_\alpha \cong \lambda\Lambda_m$. The Gram matrix of L_α is rational and that of Λ_m is primitive with a diagonal entry 2, so $\lambda = \frac{1}{2}\min L_\alpha \in \mathbb{Q}$. Then $\beta = \alpha/\lambda$ has $\mathrm{Tr}(\beta xy) \in \mathbb{Z}$ for all x, y, that is $\beta \in \mathfrak{d}^{-1}$; it has $\mathrm{Tr}(\beta x^2) \in 2\mathbb{Z}$; and $\mathrm{Nm}(\beta) D_K = \mathrm{Nm}(\beta\mathfrak{d}) = \det \Lambda_m$.

(ii)⇒(i). The lattice $(\mathcal{O}_K, \mathrm{Tr}(\beta xy))$ is even, positive and of determinant $\det \Lambda_m$. Hence its minimum is at least 2, and by Hermite's inequality at most $\gamma_m (\det \Lambda_m)^{1/m} = 2$. So it attains γ_m, and $\gamma_K^m \geq \gamma_m^m D_K$.

Now assume (ii). If 2 were unramified, $\mathcal{O}_K/2\mathcal{O}_K$ would be reduced, so squaring would be bijective on it and the squares would span $\mathcal{O}_K$ modulo $2\mathcal{O}_K$. Then $\mathrm{Tr}(\beta z) \in 2\mathbb{Z}$ for all $z \in \mathcal{O}_K$, i.e. $\beta/2 \in \mathfrak{d}^{-1}$, and 2^m would divide $\mathrm{Nm}(\beta\mathfrak{d}) = \det \Lambda_m \in \{3, 4\}$. This is impossible. The dual lattice of $(\mathcal{O}_K, \mathrm{Tr}(\beta xy))$ is $\beta^{-1}\mathfrak{d}^{-1}$, whence $\Lambda_m^{\#}/\Lambda_m \cong \beta^{-1}\mathfrak{d}^{-1}/\mathcal{O}_K \cong \mathcal{O}_K/\beta\mathfrak{d}$. For $m = 2$ this group is $\mathbb{Z}/3$, so $\beta\mathfrak{d}$ is a prime of norm 3. For $m = 3$ it is $\mathbb{Z}/4$. An integral ideal I of norm 4 with $\mathcal{O}_K/I$ cyclic has the form $\mathfrak{p}^2$ with $\mathfrak{p}$ of degree 1 and $2 \notin \mathfrak{p}^2$, i.e. $\mathfrak{p}$ unramified. As 2 ramifies, $2\mathcal{O}_K = \mathfrak{p}\mathfrak{q}^2$. □

Corollary 8.2. *Let K be real quadratic. Then $\gamma_K^2 = \frac{4}{3} D_K$ if and only if $\mathcal{O}_K$ has a $\mathbb{Z}$–basis (u, v) with*

$$\mathrm{Nm}(u) + \mathrm{Nm}(v) + \mathrm{Nm}(u - v) = 0.$$

In that case $D_K = 4(A^2 + AC + C^2)$ with $A = \mathrm{Nm}(u)$, $C = \mathrm{Nm}(v)$. In particular $D_K \equiv 12 \bmod 16$, i.e. $K = \mathbb{Q}(\sqrt{d})$ with $d \equiv 3 \bmod 4$.

Proof. For a basis (u, v) put $L_j(x, y) = \sigma_j(xu + yv)$ and $F = L_1 L_2 = Ax^2 + Bxy + Cy^2$, the norm form, of discriminant D_K. The forms $\mathrm{Tr}(\alpha(xu + yv)^2)$, $\alpha \gg 0$, are the forms $a_1 L_1^2 + a_2 L_2^2$ with $a_1, a_2 > 0$. These are exactly the positive forms $ax^2 + bxy + cy^2$ apolar to F, i.e. with $aC - \frac{1}{2}bB + cA = 0$: this condition is linear and holds for L_1^2 and L_2^2. So their roots τ in the upper half plane fill the geodesic $A|\tau|^2 + B\,\mathrm{Re}\,\tau + C = 0$. Some $\mathrm{Tr}(\alpha x^2)$ is hexagonal iff, for some basis, the root $\rho = e^{2\pi i/3}$ of $x^2 + xy + y^2$ lies on this geodesic, i.e. $A - \frac{1}{2}B + C = 0$. Since $\mathrm{Nm}(u - v) = A - B + C$, this is the stated condition. Then $D_K = B^2 - 4AC = 4(A^2 + AC + C^2)$. Since $A^2 + AC + C^2 \not\equiv 2 \bmod 4$, $D_K/4 \equiv 3 \bmod 4$. □

For example $(u, v) = (1, 2 + \sqrt{3})$ for $\mathbb{Q}(\sqrt{3})$, with norms $1, 1, -2$, and $(1, 3 + \sqrt{7})$ for $\mathbb{Q}(\sqrt{7})$, with norms $1, 2, -3$.

Computational Observation 8.3. The criteria of Theorem 8.1 and Corollary 8.2 agree with the exact values of γ_K for all 30,394 real quadratic fields with $D_K \leq 10^5$, of which 983 attain the bound. They also agree for all 4804 totally real cubic fields with $D_K \leq 10^5$, of which 306 attain it. In the cubic case the test whether $\mathfrak{p}^2\mathfrak{d}^{-1}$ has a totally positive generator used `bnfinit`, so that half of the check assumes GRH. Every attaining field has $D_K \equiv 12 \bmod 16$. In the cubic case 959 further fields have $2\mathcal{O}_K = \mathfrak{p}\mathfrak{q}^2$ without attaining the bound. The 38 cubic fields with a vertex of degree 6 attain it, as in Proposition 4.6.

Remark 8.4. The same proof works for $4 \le m \le 8$, given the classical list of critical lattices D_4, D_5, E_6, E_7, E_8. These are unique up to similarity, and their even forms have determinants $4, 4, 3, 2, 1$, all less than 2^m. In each case equality forces 2 to ramify in K.

9. Indecomposables and squares of units

In this section K is a totally real cubic field, $\mathcal{O} = \mathcal{O}_K$, $D = D_K$, $U = \mathcal{O}^\times$, U^+ the totally positive units and U^2 the squares of units. An element of $\mathcal{O}^+$ (the totally positive integers) is *indecomposable* if it is not a sum of two elements of $\mathcal{O}^+$; let Ind be the set of indecomposables. It contains 1 and is stable under multiplication by U^+. Let $\mathcal{O}^\vee = \{\beta \in K : \mathrm{Tr}(\beta\mathcal{O}) \subseteq \mathbb{Z}\}$ be the codifferent, of norm $1/D$. Let $\mathcal{K} = \mathcal{K}_{(+,+,+)}$ be the totally positive Klein polyhedron of §7. Its boundary modulo U^+ is a torus (Proposition 7.1). Every facet F is a compact lattice polygon with a functional $L_F(x) = \mathrm{Tr}(\gamma_F x)$, $\gamma_F \in K$ totally positive, such that $L_F \ge 1$ on $\mathcal{K}$ and $L_F = 1$ on F. The cone determinant of a triangle $\mathrm{conv}(v_1, v_2, v_3)$, $v_i \in \mathcal{O}$, is $[\mathcal{O} : \mathbb{Z}v_1 + \mathbb{Z}v_2 + \mathbb{Z}v_3]$.

The question is whether Ind can consist of one class modulo U^2, which is then U^2 itself, or more weakly of units only.

Lemma 9.1. (a) *Totally positive units are indecomposable.* (b) *Vertices of $\mathcal{K}$ are indecomposable.* (c) *If $x \in \mathcal{O}^+$ and $L_F(x) < 2$ for some facet F, then x is indecomposable.*

Proof. (a) If $0 < \sigma_i(y) < \sigma_i(u)$ for all i, then $0 < \mathrm{Nm}(y) < 1$. (b) If a vertex is $v = y + z$ with $y, z \in \mathcal{O}^+$, then y and $y + 2z$ lie in $\mathcal{K}$, and v is their midpoint. (c) A sum $y + z$ has $L_F(y + z) \ge 2$. □

Lemma 9.2. *Let $\alpha \in \mathcal{O}^\vee$ be totally positive with $\mathrm{Tr}(\alpha x^2) \ge 1$ for all $x \in \mathcal{O} \smallsetminus 0$, and let u_1, u_2, u_3 be units with distinct squares and $\mathrm{Tr}(\alpha u_i^2) = 1$. Then $\mathrm{Tr}(\alpha u_j u_k) = \delta_{jk}$, $\{u_1, u_2, u_3\}$ is a $\mathbb{Z}$–basis of $\mathcal{O}$, and $\mathrm{Nm}(\alpha)D = 1$, i.e. $\alpha\mathcal{O} = \mathcal{O}^\vee$. In particular two such α differ by a totally positive unit.*

Proof. The form $b(x, y) = \mathrm{Tr}(\alpha xy)$ is integral on $\mathcal{O}$, because $\alpha \in \mathcal{O}^\vee$, and its determinant is $\mathrm{Nm}(\alpha)D$, a positive integer. For $j \ne k$ we have $u_j \pm u_k \ne 0$, so $1 \le b(u_j \pm u_k, u_j \pm u_k) = 2 \pm 2b(u_j, u_k)$. Hence $|b(u_j, u_k)| \le \frac{1}{2}$, and so $b(u_j, u_k) = 0$. The u_j are orthonormal, hence independent, and $L = \sum_j \mathbb{Z}u_j$ has determinant $1 = [\mathcal{O} : L]^2 \,\mathrm{Nm}(\alpha)D$. □

In the language of §2: a perfect form that is integral (in the codifferent) and whose minimal vectors are units is the cubic lattice I_3 with an orthonormal basis of units.

Lemma 9.3. *Let $F = \mathrm{conv}(u_1^2, u_2^2, u_3^2)$ be a triangular facet of $\mathcal{K}$ with $u_i \in U$ and cone determinant 1. Then $\gamma_F\mathcal{O} = \mathcal{O}^\vee$. All such facets form a single U^+–orbit.*

Proof. $L_F = 1$ on the $\mathbb{Z}$–basis u_1^2, u_2^2, u_3^2, so $L_F(\mathcal{O}) \subseteq \mathbb{Z}$ and $\gamma_F \in \mathcal{O}^\vee$. For $x \ne 0$ the square x^2 lies in $\mathcal{O}^+ \subset \mathcal{K}$, so $\mathrm{Tr}(\gamma_F x^2) = L_F(x^2) \ge 1$. Lemma 9.2 applies. If F' is another such facet, then $\gamma_{F'} = \eta\gamma_F$ with $\eta \in U^+$. Since $\mathcal{K}$ is U^+–invariant, the facet with functional $\mathrm{Tr}(\eta\gamma_F\,\cdot)$ is $\eta^{-1}F$; so $F' = \eta^{-1}F$. □

Theorem 9.4. *In every totally real cubic field some indecomposable is not the square of a unit. Equivalently,* Ind *has at least two classes modulo multiplication by U^2.*

Proof. Suppose $\mathrm{Ind} = U^2$.

(1) $U^+ = U^2$, by Lemma 9.1(a).
(2) The vertices of $\mathcal{K}$ are units, by Lemma 9.1(b). They form a non–empty U^+–invariant subset of U^+, so they are all of U^+ and $V = 1$ modulo U^+.
(3) A facet contains no lattice point besides its vertices. Such a point p would lie in $\mathcal{O}^+$ with $L_F(p) = 1$, hence be indecomposable, and it is not a unit, since the units are vertices.

(4) Let F_k be the number of facet classes with k vertices. Euler's formula $V - E + F = 0$ with $V = 1$ and $2E = \sum_k kF_k$ gives $\sum_k (k-2)F_k = 2$. So there are two triangle classes or one quadrilateral class.
(5) There is no quadrilateral. By (3) it would be an empty lattice quadrilateral in the lattice plane $\{L_F = 1\}$, hence a parallelogram. By Pick's theorem it has area 1, the two triangles cut off by a diagonal are unimodular, and with $p_2 = (0,0)$, $p_1 = (1,0)$, $p_3 = (0,1)$ the fourth vertex is $(1+k, 1-k)$, convex only for $k = 0$. On the torus with $V = 1$, $E = 2$, $F = 1$ the sides are glued as $aba^{-1}b^{-1}$, since the torus is orientable and the totally positive units preserve orientation. So the vertices are $1, a, ab, b$ in cyclic order, with $a, b \in U^+ \smallsetminus \{1\}$. The parallelogram condition $1 + ab = a + b$ gives $(1-a)(1-b) = 0$, which is impossible.
(6) Both triangles have cone determinant 1. Otherwise the half–open parallelepiped $\{\sum \lambda_i v_i : 0 \le \lambda_i < 1\}$ of a triangle $F = \mathrm{conv}(v_1, v_2, v_3)$ contains a lattice point $x \neq 0$. It lies in $\mathcal{O}^+$, and so does $x' = v_1 + v_2 + v_3 - x$, so $1 \le L_F(x) = 3 - L_F(x') \le 2$. If $L_F(x) > 1$ then x is not on the sail, hence not a unit, and if moreover $L_F(x) < 2$ it is indecomposable by Lemma 9.1(c). If $L_F(x) = 1$, then x is a lattice point of F other than a vertex, and if $L_F(x) = 2$, then x' is one. Both contradict (3).
(7) By (1), (2) and (6) both triangle classes satisfy Lemma 9.3. So they form one U^+–orbit, contradicting (4). □

Remark 9.5. (a) The perfect forms enter through Lemma 9.2. Under the hypothesis of the proof the squared sail and $\mathcal{K}$ coincide, since both are $\mathrm{conv}(U^+) + \mathbb{R}^3_{\ge 0}$. So each facet is a perfect form, and Lemma 9.2 identifies it with the integral form I_3. Two classes of such forms cannot coexist. (b) The real quadratic analogue of Theorem 9.4 is false: $\mathrm{Ind} = U^2$ for $\mathbb{Q}(\sqrt{5})$. There the sail modulo U^+ is a circle, made of a single edge. (c) If every indecomposable is a unit, steps (2)–(6) still hold with U^+ in place of U^2. The sail then consists of two unimodular triangles $\mathrm{conv}(1, a, ab)$ and $\mathrm{conv}(1, b, ab)$ with $a, b \in U^+$. From $\mathcal{O} = \mathbb{Z} + \mathbb{Z}a + \mathbb{Z}ab = \mathbb{Z} + \mathbb{Z}b + \mathbb{Z}ab$ and the integrality of L_F one gets $b = -a + m + n\,ab$ with $m + n \ge 3$, and Theorem 9.4 adds $U^+ \neq U^2$. We do not know whether this configuration occurs. It does not for $D \le 10^6$ (Observation 9.7). (d) In degree $m \ge 4$, Lemmas 9.2 and 9.3 hold verbatim, with Gram matrix I_m. Steps (4)–(6) do not: empty lattice simplices of dimension at least 3 need not be unimodular, the parallelepiped only gives $1 \le L_F(x) \le m - 1$, and a torus of dimension at least 3 can have one cell.

Lemma 9.6. *Let ε be a unit with $\sigma_1(\varepsilon) > 1$, $0 < \sigma_2(\varepsilon) < 1$ and $-1 < \sigma_3(\varepsilon) < 0$, such that $1 - \varepsilon$ is also a unit. Then ε is a root of $P_t(x-1)$, where $P_t(x) = x^3 - tx^2 - (t+3)x - 1$ and $t = \mathrm{Tr}(\varepsilon) - 3$. So K is the simplest cubic field K_t and $\mathbb{Z}[\theta_t] \subseteq \mathcal{O}$ with $\theta_t = \varepsilon - 1$.*

Proof. Let $x^3 - Ax^2 + Bx - C$ be the minimal polynomial of ε. Then $C = \mathrm{Nm}(\varepsilon) < 0$, so $C = -1$. Its value at 1 is $\mathrm{Nm}(1-\varepsilon) = \prod_i (1 - \sigma_i(\varepsilon)) < 0$, so $2 - A + B = -1$. With $t = A - 3$ this polynomial is $x^3 - (t+3)x^2 + tx + 1 = P_t(x-1)$. □

Computational Observation 9.7. These are certified computations.

(a) Among the 4804 fields with $D \le 10^5$, exactly 15 have an integral perfect form with unit minimal vectors (Lemma 9.2). There is one class in each, and the fields are the simplest cubic fields K_t with $\mathcal{O}_K = \mathbb{Z}[\theta_t]$ and $D_K \le 10^5$, including $t = 0$.
(b) For $D \le 10^6$, $\ell_{\mathcal{O}_K} = 1$ holds exactly for the 26 simplest cubic fields with $\mathcal{O}_K = \mathbb{Z}[\theta_t]$, $t = -1$ and $1 \le t \le 29$. In each of them the neighbour $\varepsilon = N_k(1)$ of 1 in each direction k is a unit with the sign pattern of Lemma 9.6, up to a permutation of the places, and $1 - \varepsilon$ is a unit adjacent to 1. For $t = 0$ ($D = 81$) the neighbours $N_k(1)$ behave in the same way, but $\ell = 2$.
(c) Sails data (pav10/sails): the total cone determinant S^+ of the totally positive sail modulo U^+ is at least 3 for every field with $D \le 10^6$. So no such field has only units as indecomposables.

Conjecture 9.8. *$\ell_{\mathcal{O}_K} = 1$ if and only if $K = K_t$ is a simplest cubic field with $\mathcal{O}_K = \mathbb{Z}[\theta_t]$ and $t \neq 0, -3$.*

By Lemma 9.6, the "only if" part, apart from the equality $\mathcal{O}_K = \mathbb{Z}[\theta_t]$, would follow from this statement: if $\ell_{\mathcal{O}_K} = 1$, then for some direction k the neighbour $\varepsilon = N_k(1)$, normalised by $\sigma_k(\varepsilon) > 0$, has one positive and one negative small conjugate, and $1 - \varepsilon$ is a relative minimum. Both small conjugates cannot be positive, since then $\varepsilon - 1$ would contradict the minimality of ε. In the census both conditions hold in every direction (Observation 9.7(b)), and $\{1, \varepsilon, 1-\varepsilon\}$ is the degenerate minimal triple of Lemma 7.8.

10. Further directions

10.1. **Other degrees.** The proofs of Lemmas 6.1–6.4 and of Theorem 6.5 use only the dimension of the log–unit torus. For a totally real field of degree m they give

$$\frac{2^{m-1}\operatorname{Reg}_K}{(\log \gamma_m^m D_K)^{m-1}} \;\le\; \mu_K \;\le\; \ell_{\mathcal{O}_K} \;\le\; \frac{2^{m-1}\operatorname{Reg}_K}{(\log 2)^{m-1}}. \tag{5}$$

Here the range ball of radius ρ in H_0 has volume $\sqrt{m}\,\rho^{m-1}$, there are 2^{m-1} sign classes, and γ_m is Hermite's constant.

Proposition 10.1 (General signature)**.** *Let K be any number field with $d = r_1 + r_2$ and $\mathfrak{o}$ an order. Then $\partial K_1(m_{\mathfrak{o}})/\Gamma_{\mathfrak{o}}$ is a closed aspherical $(d-1)$-manifold with fundamental group $\mathbb{Z}^{d-1}$. It carries a finite regular CW structure whose face numbers $f_0 = n_K, f_1, \dots, f_{d-1} = \mu_K$ satisfy $\sum_i (-1)^i f_i = \chi(T^{d-1}) = 0$. If K is totally real of degree m, then moreover*

$$n_K \;\ge\; \frac{2m}{\tau_m}\,\mu_K,$$

where τ_m is the lattice kissing number in dimension m.

Proof. The first part is the proof of Theorem 4.3 with 2 replaced by $d-1$. For the inequality, count incidences between vertices and facets. Each facet of $K_1(m_{\mathfrak{o}})$ is an $(m-1)$–polytope and so has at least m vertices. Each vertex lies on as many facets as it has extreme minimal vectors up to sign. The form $x \mapsto \operatorname{Tr}(\alpha x^2)$ is a positive definite quadratic form on $\mathfrak{o} \cong \mathbb{Z}^m$, so it has at most $\tau_m/2$ minimal vectors up to sign. Hence $m\,\mu_K \le (\tau_m/2)\,n_K$. □

For $m = 3$ one has $\tau_3 = 12$, and this recovers $n_K \ge \frac{1}{2}\mu_K$. Consequently $n_K \gg_m \operatorname{Reg}_K/(\log D_K)^{m-1}$ in every degree. Only the upper bound $n_K \le 2\mu_K$ uses the surface case $d = 3$. For $d \ge 4$ an upper bound for n_K in terms of μ_K would need the Dehn–Sommerville relations of the torus complex together with a bound on the number of vertices per facet.

10.2. **Relation to indecomposables.** The extreme squares are a non–linear cousin of the additively indecomposable totally positive integers, whose sail is studied by Kala–Man [11] and Kala–Sgallová–Tinková [12]; see also Karpenkov [13] for Klein sails. The set $q(\mathfrak{o})$ is not a lattice. By Lemma 6.3, however, the squared sail is controlled by the relative minima of the genuine lattice $\iota(\mathfrak{o})$. This is Voronoi's multidimensional continued fraction, not the Jacobi–Perron algorithm, and it is the reason why lattice methods do apply, contrary to what an earlier draft suggested. Section 7 makes the link explicit: the relative minima are the Klein sail vertices together with German's points [4, 9]. The totally positive ones are indecomposable (Lemma 7.6). Their cell structure is the Minkowski–Voronoi complex of Karpenkov–Ustinov [14]. Section 9 uses perfect forms in the other direction, to constrain the indecomposables themselves.

10.3. **Open problems.**

(1) Prove unconditionally that $\sup n_K = \infty$ over maximal orders of totally real cubic fields. By Corollary 6.8 it suffices to find fields with $\mathrm{Reg}_K/(\log D_K)^2$ unbounded. Alternatively, one could find a parametric family on which the Voronoi output has visible parametric structure with μ_K growing, as in Yasaki's explicit long–period quadratic families. The explicit–unit families $x(x-a)(x-2a)-1$ and $x(x-a)(x+a+1)-1$ that we tested stay on or near the floor; when $\mathrm{Reg}_K \asymp (\log D_K)^2$ the lower bound of Theorem 6.5 is $O(1)$ and forces nothing. A successful family presumably needs fundamental units that are long products of elements of small norm.

(2) Determine $\liminf \mu_K/\mathrm{Reg}_K$ in distribution, if it exists. The median over $h_K = 1$ falls from 0.62 for $D \le 20000$ to 0.55 for $8\cdot 10^4 < D \le 10^5$, and the three large fields above give about 0.47. This is the cubic analogue of Lévy's constant for continued fractions; for real quadratic fields the census gives ℓ_d/Reg with median 0.841 over the fields with $D \le 10^5$ and $\mathrm{Reg} > 20$, against Lévy's $12\log 2/\pi^2 = 0.8428$.

(3) Prove $\mu_K = 1$ for all Ennola fields with $a \ge 4$ and all Shanks fields.

(4) Prove a lower bound $n_K \ge c\,\ell_{\mathfrak{o}}$ with an absolute $c > 0$, the cubic analogue of $h_d \ge \ell_d/2$ [27, Thm. 5.1]. The census suggests $c = \frac{4}{9}$ (Observation 5.7). In all 832 fields with $D \le 20000$ every relative minimum is an extreme square or adjacent to one; that is, the extreme squares dominate the adjacency graph of the relative minima. This is a cubic form of Yasaki's observation that the minimal vectors of consecutive perfect forms never skip two consecutive convergents. The degrees in that graph are not bounded in the data (they reach 26 in a sample of fields with $D \le 20000$), so this does not by itself give a linear bound. The stronger statement that every minimal triple contains an extreme square fails, for 135 of the 50,176 minimal triples in this range.

(5) Exclude the two configurations of Proposition 4.6, i.e. prove $n_K \ge 2$; and explain why $n_K = 3$ does not occur for $D \le 10^5$.

(6) Can an extreme square be a German point with $k \ge 3$, or lie on a facet at integer distance greater than 1? Relative minima of both kinds occur (Observation 7.9).

(7) Is there a totally real cubic field in which every indecomposable is a unit? By Remark 9.5(c) its totally positive sail would be two unimodular triangles $\mathrm{conv}(1, a, ab)$, $\mathrm{conv}(1, b, ab)$ with $b = -a + m + n\,ab$, $m + n \ge 3$, and $U^+ \ne U^2$.

(8) Prove Conjecture 9.8. By Lemma 9.6 the main step is to show that $\ell_{\mathcal{O}_K} = 1$ forces $1 - N_k(1)$ to be a relative minimum.

Computational note

The code (PARI/GP, file `voronoi_unary.gp`) implements §5.2; every value reported in §5 passed both certificates. An independent reimplementation for totally real fields of any degree (Python on PARI, repository `pav10/perfect_forms`) reproduces all 832 census records (counts, vertex degrees and norms of the extreme squares) and the three large fields of §5, with both certificates. It produced Observation 5.7 and the data of §§7–8; the sails were computed with the code of the repository `pav10/sails`. As a test in degree 2, it gives exactly Yasaki's total of 10,732,735 perfect unary forms over the 121,580 fields $\mathbb{Q}(\sqrt{d})$, $d \le 200000$ squarefree [27, §4], and confirms [27, Thm. 5.1] and [18, Thm. 1.2] in all of them. The field lists were cross–checked against the LMFDB (`https://www.lmfdb.org`).

References

[1] R. Brauer, *On the zeta–functions of algebraic number fields*, Amer. J. Math. **69** (1947), 243–250.

[2] J. Buchmann, *Abschätzung der Periodenlänge einer verallgemeinerten Kettenbruchentwicklung*, J. Reine Angew. Math. **361** (1985), 27–34.
[3] J. Buchmann, *On the period length of the generalized Lagrange algorithm*, J. Number Theory **26** (1987), 31–37.
[4] V. A. Bykovskii, *Local minima of lattices and vertices of Klein polyhedra*, Funct. Anal. Appl. **40** (2006), 56–57.
[5] H. Cohen, *A Course in Computational Algebraic Number Theory*, Grad. Texts in Math. **138**, Springer, Berlin, 1993.
[6] H. Cohen and J. Martinet, *Étude heuristique des groupes de classes des corps de nombres*, J. Reine Angew. Math. **404** (1990), 39–76.
[7] B. N. Delone and D. K. Faddeev, *The Theory of Irrationalities of the Third Degree*, Transl. Math. Monographs **10**, Amer. Math. Soc., Providence, RI, 1964.
[8] V. Ennola, *Cubic number fields with exceptional units*, in: Computational Number Theory (Debrecen, 1989), de Gruyter, Berlin, 1991, pp. 103–128.
[9] O. N. German, *Klein polyhedra and relative minima of lattices*, Math. Notes **79** (2006), 505–510.
[10] P. E. Gunnells and D. Yasaki, *Perfect forms over totally real number fields*, arXiv:0908.3130.
[11] V. Kala and S. H. Man, *Sails for universal quadratic forms*, Selecta Math. (N.S.) **31** (2025), Paper No. 26.
[12] V. Kala, E. Sgallová and M. Tinková, *Arithmetic of cubic number fields: Jacobi–Perron, Pythagoras, and indecomposables*, J. Number Theory **273** (2025), 37–95.
[13] O. Karpenkov, *Geometry of Continued Fractions*, Algorithms and Computation in Mathematics, vol. 26, Springer, Berlin, 2013.
[14] O. Karpenkov and A. Ustinov, *Geometry and combinatorics of Minkowski–Voronoi 3–dimensional continued fractions*, J. Number Theory **176** (2017), 375–419.
[15] A. Leibak, *On additive generalization of Voronoï's theory for algebraic number fields*, Proc. Estonian Acad. Sci. Phys. Math. **54** (2005), 195–211.
[16] A. Leibak, *On additive generalisation of Voronoï's theory of perfect forms over algebraic number fields*, PhD thesis, Tallinn University of Technology, 2006.
[17] H. Komatsu and T. Watanabe, *Perfect unary forms over certain cubic fields*, Int. J. Number Theory **10** (2014), no. 5, 1337–1342.
[18] A. Leibak, C. Porter and C. Ling, *Unit reducible fields and perfect unary forms*, J. Théor. Nombres Bordeaux **35** (2023), 867–895.
[19] F. Lemmermeyer and A. Pethő, *Simplest cubic fields*, Manuscripta Math. **88** (1995), 53–58.
[20] K. Okuda and S. Yano, *A generalization of Voronoï's theorem to algebraic lattices*, J. Théor. Nombres Bordeaux **22** (2010), 727–740.
[21] C. Porter and A. Mendelsohn, *An upper bound on the number of classes of perfect unary forms in totally real number fields*, Int. J. Number Theory (2024), doi:10.1142/S1793042124501008; arXiv:2208.03304.
[22] S. S. Ryshkov, *The polyhedron $\mu(m)$ and some extremal problems of the geometry of numbers*, Soviet Math. Dokl. **11** (1970), 1240–1244.
[23] D. Shanks, *The simplest cubic fields*, Math. Comp. **28** (1974), 1137–1152.
[24] G. F. Voronoï, *On a generalization of the algorithm of continued fractions* (Russian), Doctoral dissertation, Warsaw, 1896.
[25] G. Voronoï, *Nouvelles applications des paramètres continus à la théorie des formes quadratiques. Premier mémoire*, J. Reine Angew. Math. **133** (1908), 97–178.
[26] T. Watanabe, S. Yano and T. Hayashi, *Voronoï's reduction theory of GL_n over a totally real number field*, in: Diophantine Methods, Lattices, and Arithmetic Theory of Quadratic Forms, Contemp. Math. **587**, Amer. Math. Soc., Providence, RI, 2013, pp. 213–232.
[27] D. Yasaki, *Perfect unary forms over real quadratic fields*, J. Théor. Nombres Bordeaux **25** (2013), 759–775.